Bachelor Mathematics

Track: Analysis and Dynamical Systems

Bachelor Thesis

Diophantine Approximation for Solving Thue Equations

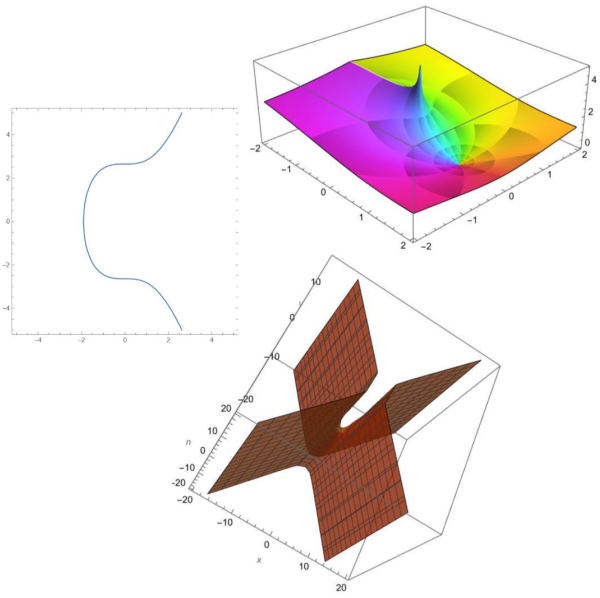
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June 29, 2024

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Abstract

A Thue Equation is a Diophantine equation of the form $F(X, Y) = m$, with m - a given nonzero integer and F - a binary form of degree $n > 2$, having integer coefficients and irreducible over $\mathbb{Q}$. A result, due to the Norwegian mathematician Axel Thue, states that a Thue equation has only finitely many solutions. The aim of this thesis is to derive a practical method for finding all solutions to a general Thue equation and then apply it on concrete examples. In particular, a method, due to Tzanakis and de Weger, for solving Thue equations, will be presented in detail. This method will be used in practice to solve the Diophantine equation

$$4X^4 - 3Y^4 = \pm 61, \quad X, Y \in \mathbb{Z}.$$

Additional examples of Thue equations and their solutions obtained with the suggested method will also be presented. In their attempt to solve the Diophantine equation

$$F(X, Y) = m, \quad m \in \mathbb{Z}\backslash\{0\},$$

Tzanakis and de Weger first transform it into an equation inside a number field. Then, invoking the Baker-Wüstholz Theorem for logarithmic forms, they are in the position to pass the problem of determining a bound on the exponents of the units from the representation of the candidate solution, to the problem of how small a linear form in complex logarithms can become. In this situation, techniques from lattice theory allow us to significantly reduce the number of possible solutions. In turn, we are able to solve our problem by direct enumeration.

Certain choices in the method of Tzanakis and de Weger can have an effect on the performance of the method. In this thesis, a discussion on how the choice of fundamental units can affect the reduction step of the method, will be presented, with suggestions for future consideration.

Another method for solving Thue equations is found in the work of Yuri Bilu and Guillaume Hanrot. Its reduction step follows a more efficient manner in which one approximates an inhomogeneous linear form using only two-dimensional lattices. The method of Bilu and Hanrot further allows for more efficient final enumeration, which makes it a powerful tool for solving Thue equations of very high degree.

This thesis ends with a short summary and a conclusion on the obtained results.

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1 Introduction

The term *Diophantine Equation* refers to a polynomial equation with integer coefficients, where solutions are sought in the integers $\mathbb{Z}$ or the rationals $\mathbb{Q}$, with possibly further conditions on the unknowns. These types of equations have fascinated mathematicians for millennia. They are named after Diophantus of Alexandria (AD 214 to 284 or 298), the "father of algebra" [1], who is most notably known for his work *Arithmetica* - one of the most influential books in the history of mathematics. *Arithmetica* is a collection of 150 problems of equations up to third degree, some of which deal with the theory of numbers. Famous examples of Diophantine equations studied by mathematicians include the *Hardy-Ramanujan Equation*:

$$w^3 + x^3 = y^3 + z^3, \quad w, x, y, z \in \mathbb{N}, \quad w \neq y \quad \text{and} \quad w \neq z.$$

This equation, central to the concept of *taxicab numbers*, describes sums of cubes in essentially two different ways. Another example is the equation stated in *Fermat's Last Theorem*:

$$x^n + y^n = z^n, \quad x, y, z \in \mathbb{Z}, \quad n \in \mathbb{N}.$$

Sir Andrew Wiles famously proved in 1994 that this equation has no integer solutions for $n \geq 3$, apart from the triple $(x, y, z) = (0, 0, 0)$. The Pell's equation is another example:

$$x^2 - Ny^2 = \pm 1, \quad x, y \in \mathbb{Z} \quad \text{and } N \in \mathbb{N} \text{ is not a square.}$$

The case when $N = 2$ was studied as early as 400 BC in both India and Ancient Greece. Other examples of Diophantine equations can be found in many classical number theory textbooks. Nowadays, Diophantine equations are a very active area of mathematical research, with many applications, ranging from Physics, Geometry of Numbers and Room Acoustics. More details on the practical applications of Diophantine equations can be found in [3].

Despite their intriguing nature, Diophantine problems share one intrinsic characteristic: they are easy to formulate but often extremely difficult to solve. Various solution techniques from elementary number theory involve clever observations, factorization arguments, local obstructions modulo prime powers p^n, continued fraction expansions, or determining bounds on the possible solution set, to name a few. A good introduction to these techniques can be found in [1] and Chapter 7 of [3]. While elementary methods have a limited scope, strategies from algebraic and analytic number theory allow us to tackle a larger class of problems, though they apply only to specific forms of equations, such as Mordell Equations, Pell-type, or ones represented by quadratic binary forms. For a comprehensive treatment of degree 2 Diophantine equations see [2].

The quest for a unified theory of solving polynomial equations over the integers drew the attention of the great polymath David Hilbert. In his address at the first International Mathematical Congress, held in Paris in 1900, Hilbert formulated a list of 10 (and later expanded to 23) problems to challenge mathematicians in the 20th century. Among them was the problem of devising a general algorithm for deciding on the existence of solutions to a given Diophantine equation, known within mathematical folklore as *Hilbert's Tenth Problem* [6]. However, in 1970, Yuri Matiyasevich proved in his doctoral dissertation that a general algorithm for solving all Diophantine equations cannot exist. And, although we cannot hope for a universal procedure, we can still study "sufficiently large" classes of equations, such as those of the form:

$$F(X, Y) = m,$$

where $F(X, Y) \in \mathbb{Z}[X, Y]$ is a binary form of degree $n \geq 3$, m is a fixed nonzero integer, and $F(X, Y) \in \mathbb{Z}[X, Y]$ is irreducible in $\mathbb{Q}[X, Y]$. These equations are known as Thue equations, named after the Norwegian mathematician Axel Thue. In 1909, Thue demonstrated that such an equation has only finitely many solutions over the integers. The main weakness of Thue's result was that it was *ineffective*, in the sense that it provided no way of determining the actual solutions to an equation of this type (for more details, check Bombieri's account in [12]). It wasn't until the late 1960s, with the groundbreaking results of Alan Baker, that mathematicians obtained a theoretical algorithm for finding all solutions to a Thue equation. Two decades later, combining computational Diophantine approximation techniques from lattice theory and the LLL-reduction algorithm with Baker's results, Tzanakis and de Weger developed a practical algorithm for solving Thue equations. Since then more methods were developed and a large number of Diophantine equations have been computationally solved.

Thue equations find several theoretical applications. They can be related to the study of determining integral points on elliptic curves and they also give rise to generalizations such as the so-called Thue-Mahler equations. Modern research continues to build on Thue's work, using advanced techniques from Transcendental Number Theory, Computational Number Theory, and Algebraic Geometry to find explicit solutions to equations of this type (see [12]), portraying Thue-type equations as intriguing objects worthy of our attention.

The primary goal of this thesis is to derive and apply the method of Tzanakis and de Weger for solving Thue equations. Our main source of literature is the book by Nigel Smart [4]. In Chapter 2 we give a short introduction to the theory of lattices. Chapter 3 presents one application of LLL-reduction in approximating complex linear forms. In Chapter 4 we give a detailed derivation of the method of Tzanakis and de Weger and its application on a concrete example. Additional Thue equations will be presented with a link to our `Mathematica` implementations. Chapter 4 ends with a discussion on the efficiency of the method. Chapter 5 is devoted to another method for Thue equations due to Yuri Bilu and Guillaume Hanrot. We will present a brief derivation with a comment on how this second method provides a computational advantage over the one presented in Chapter 4. We will close this thesis with a final conclusion on the obtained results and a comment on possible future considerations.

2 Basic Theory of Lattices

This chapter is devoted to lattices. Here we present the main definitions and introduce the notion of LLL-reduction. We also describe the process of associating an approximation lattice to a linear form. Key references for the current chapter include the textbooks [4] and [16], together with the lecture notes [7].

2.1 Introduction to Lattices

We start our discussion with some of the basics of lattice theory. First, let us introduce the main definitions:

Definition 2.1 (Lattice)**.** A *lattice* $\mathcal{L}$ is a subgroup of $\mathbb{R}^m$, such that there exist n linearly independent vectors $\vec{b_1}, ..., \vec{b_n} \in \mathbb{R}^m$ with the property that $\mathcal{L} = \left\{ \sum_{i=1}^{n} x_i \vec{b_i} \mid x_i \in \mathbb{Z} \right\}$. We refer to $\vec{b_1}, ..., \vec{b_n}$ as a basis for the lattice. If we define B to be the $m \times n$ matrix with columns $\vec{b_i}$, then the lattice generated by B is $\mathcal{L}(B) \stackrel{\text{def}}{=\!=} \mathcal{L}(\vec{b_1}, ..., \vec{b_n}) \stackrel{\text{def}}{=\!=} \{B\vec{x} \mid \vec{x} \in \mathbb{Z}^n\}$. We refer to m as the dimension of the lattice and n - the rank of the lattice. If $m = n$, the lattice is said to be full rank.

In what follows we will consider only full rank lattices, since they will serve the needs of our applications. With the above definition in mind, a natural question to ask is whether two bases B_1 and B_2 can generate the same lattice, i.e., $\mathcal{L}(B_1) = \mathcal{L}(B_2)$. To deliver an answer, we introduce the following

Definition 2.2. A matrix $U \in \mathbb{Z}^{n \times n}$ is called *unimodular* if $det(U) = \pm 1$.

One can show that the inverse U^{-1} of a unimodualr matrix U is also unimodular, and in particular, $U^{-1} \in \mathbb{Z}^{n \times n}$. We can also show that two lattice bases B_1 and B_2 are equivalent (i.e., they generate the same lattice) if there exists $U \in \mathbb{Z}^{n \times n}$ - unimodular, with $B_1 = B_2 U$. We now define the *determinant* of a lattice $\mathcal{L} = \mathcal{L}(B)$ to be $\Delta(\mathcal{L}) = |det(B)|$. Then it is immediate that $\Delta(\mathcal{L})$ is an invariant of the lattice $\mathcal{L}$, independent of the choice of basis: $\Delta(\mathcal{L}) = |det(B_1)| = |det(B_2)| \cdot |det(U)| = |det(B_2)|$, given $U \in \mathbb{Z}^{n \times n}$, unimodular.

One important parameter of every lattice is the shortest possible distance between two distinct lattice vectors, called *minimum distance*, or *first successive minimum*.

Definition 2.3 (First Successive Minimum)**.** For a lattice $\mathcal{L}$ define the *first successive minimum* $M_1 = M_1(\mathcal{L}) \stackrel{\text{def}}{=\!=} \inf \{\|\vec{x} - \vec{y}\| \mid \vec{x}, \vec{y} \in \mathcal{L},\ \vec{x} \neq \vec{y}\}$.

Since a lattice is an additive subgroup of $\mathbb{R}^n$, we can rephrase the above definition as follows: $M_1(\mathcal{L}) = \inf\left\{\|\vec{x}\| \mid \vec{x} \in \mathcal{L}\backslash\{\vec{0}\}\right\}$. A standard argument, using n-dimensional volumes of Euclidean balls in $\mathbb{R}^n$, allows us to conclude that the first successive minimum M_1 is always achieved by a nonzero lattice vector. In the context of solving Thue equations, knowing the value of M_1 will allow us to approximate a linear form using a lattice, the usefulness of which will become apparent in Chapter 4. At the same time, determining the value of M_1, known as the *shortest vector problem* (see [16]), is a very difficult computational task. The best known exact algorithms for computing the first successive minimum are rather inefficient, with a running time at least exponential in the dimension of the lattice. For an algorithm with $O(n^n)$ arithmetic operations, where n denotes the dimension of the lattice, we suggest referring to [13]. Instead of computing the exact value of the M_1, we will be satisfied with having a good enough approximation of it. This result will come in Chapter 3 in the form of one application of the LLL-reduction algorithm.

A result one usually sees at the beginning of their journey in Linear Algebra is the process of altering a basis for $\mathbb{R}^n$ to produce an *orthogonal/orthonormal* basis via the Gram-Schmidt process.

Theorem 2.4 (The Gram-Schmidt Process)**.** *Let an inner product space have a basis $\vec{b_1}, ..., \vec{b_n}$, and denote the inner product by $\langle\,,\rangle$. Then there exists an orthogonal basis $\vec{b_1^*}, ..., \vec{b_n^*}$, such that*

$$\vec{b_i^*} = \vec{b_i} - \sum_{j=1}^{i-1} \mu_{i,j}\vec{b_j^*}, \quad i = 1, ..., n,$$

where the constants $\mu_{i,j}$ are defined by $\mu_{i,j} \overset{\text{def}}{=\!=} \dfrac{\langle \vec{b_i}\ ,\ \vec{b_j^*}\rangle}{\langle \vec{b_j^*}\ ,\ \vec{b_j^*}\rangle}$.

Some immediate remarks on the Gram-Schmidt orthogonalization are that $\langle \vec{b_i^*}\ ,\ \vec{b_j^*}\rangle = 0$ for $i \neq j$; $Span(\vec{b_1}, ..., \vec{b_i}) = Span(\vec{b_1^*}, ..., \vec{b_i^*})$ for $1 \leq i \leq n$; note that the vectors $\vec{b_i^*}$ need not be a basis of $\mathcal{L}(\vec{b_1}, ..., \vec{b_n})$, in fact they need not even lie inside $\mathcal{L}(\vec{b_1}, ..., \vec{b_n})$, since the constants $\mu_{i,j}$ are not guaranteed to be in $\mathbb{Z}$. Orthogonal bases are a powerful tool in the context of Hilbert spaces, such as $\mathbb{R}^n$, and while the Gram-Schmidt process cannot directly produce lattice bases, one can still aspire to find a lattice basis that exhibits similar properties to those of a Gram-Schmidt basis. This idea can be realized in view of the following:

Definition 2.5 (LLL-reduced Basis)**.** A lattice basis B is called *LLL-reduced* (or simply *reduced*) if its associated Gram-Schmidt basis B^* and constants $\mu_{i,j}$, obtained from the Gram-Schmidt process, satisfy the two conditions:

1. $|\mu_{i,j}| \leq \frac{1}{2}, \quad 1 \leq j < i \leq n$, and
2. $\|\vec{b_i^*} + \mu_{i,i-1}\vec{b_{i-1}^*}\|^2 \geq \frac{3}{4}\|\vec{b_{i-1}^*}\|^2, \quad 1 < i \leq n.$

We can rewrite the second condition of the definition (known as the *Lovász condition*) as follows:

$$\begin{aligned}
&\|\vec{b_i^*} + \mu_{i,i-1}\vec{b_{i-1}^*}\|^2 = \langle \vec{b_i^*} + \mu_{i,i-1}\vec{b_{i-1}^*} \ , \ \vec{b_i^*} + \mu_{i,i-1}\vec{b_{i-1}^*} \rangle = \\
&= \langle \vec{b_i^*} \ , \ \vec{b_i^*} \rangle + \mu_{i,i-1}\langle \vec{b_i^*} \ , \ \vec{b_{i-1}^*} \rangle + \mu_{i,i-1}\langle \vec{b_{i-1}^*} \ , \ \vec{b_i^*} \rangle + \mu_{i,i-1}^2 \langle \vec{b_{i-1}^*} \ , \ \vec{b_{i-1}^*} \rangle = \\
&= \|\vec{b_i^*}\|^2 + \mu_{i,i-1}^2 \|\vec{b_{i-1}^*}\|^2 \geq \frac{3}{4}\|\vec{b_{i-1}^*}\|^2,
\end{aligned}$$

and rearrange the last inequality to get

$$\|\vec{b_i^*}\|^2 \geq \left(\frac{3}{4} - \mu_{i,i-1}^2\right) \|\vec{b_{i-1}^*}\|^2. \tag{2.1}$$

Inequality (2.1) will be used in Chapter 3 where we develop techniques from Lattice Approximation.

An LLL-reduced basis always exists and can be computed explicitly. The algorithms of Lenstra, Lenstra and Lovász, and de Weger's can be found in Section V.4 of [4]. There are various modifications of these algorithms and most algebra computer packages have an algorithmic implementation for computing LLL-reduced bases. For the needs of this thesis, we use `Wolfram Mathematica`. We shall not spend time on the available algorithms for lattice reduction, but rather focus on its applications for approximating linear forms in the next chapter.

2.2 Approximation Lattices

In their attempt to solve the general Thue equation, Tzanakis and de Weger reduce the classical Diophantine problem to the problem of how small a linear form in complex logarithms of algebraic numbers can become. The strategy to tackle this situation is due to de Weger. This will be the crucial step which later will allow us to solve our Diophantine problem by direct enumeration. Before we delve further into the details of this technique, we need to understand the idea behind approximating a linear form via a lattice. In this section we outline the setup of this procedure. In the next chapter we will formally justify the choices made in the following construction.

Definition 2.6 (Linear form)**.** An expression of the form

$$L(\vec{x}) = \alpha_0 + \alpha_1 x_1 + \cdots + \alpha_n x_n, \quad \alpha_i \in \mathbb{R}$$

is called a *linear form* in the variable $\vec{x} \in \mathbb{Z}^n$. A linear form is said to be *homogeneous* if $\alpha_0 = 0$ and *inhomogeneous* otherwise.

In our study, we will be interested in the case where a form is evaluated at a vector $\vec{x} \in \mathbb{Z}^n$, the components of which are bounded by some positive constants $|x_i| \leq X_i$, $1 \leq i \leq n$. Suppose now that a form $L(\vec{x}) = \alpha_0 + \alpha_1 x_1 + \cdots + \alpha_n x_n,\ \alpha_i \in \mathbb{R},$ is given and define the following matrix A:

$$A = \begin{pmatrix} 1 & & & 0 \\ & \ddots & & \\ 0 & & 1 & 0 \\ \alpha_1 & \dots & \alpha_{n-1} & \alpha_n \end{pmatrix} \in M_{n\times n}(\mathbb{R}).$$

Further, let

$$\vec{y} = (0, \dots, 0, -\alpha_0)^t \in \mathbb{R}^n.$$

Definition 2.7 (Approximation lattice)**.** The set $\mathcal{L}(A) = \{A\vec{x} : \vec{x} \in \mathbb{Z}^n\}$, with A as defined above, is called an *approximation lattice* in n-dimensional Euclidean space, associated to the form $L(\vec{x})$.

In Chapter 4 we will face the task of determining an upper bound on $A \in \mathbb{N}$ from an inequality of the form

$$|L(\vec{x})| \leq c_2 e^{-c_3 A},$$

where c_2 and c_3 are fixed positive constants. Formulated without much rigour, this amounts to determining a threshold value for the evaluations $|L(\vec{x})|$ with $\vec{x} \in \mathbb{Z}^n$, after which the absolute value of the form can not become any smaller. The strategy to tackle this situation amounts to reducing the above inequality to a problem about lattices. Let us first compare the vectors $A\vec{x}$ and $\vec{y}$:

$$A\vec{x} = \begin{pmatrix} 1 & & & 0 \\ & \ddots & & \\ 0 & & 1 & 0 \\ \alpha_1 & \dots & \alpha_{n-1} & \alpha_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ L(\vec{x}) - \alpha_o \end{pmatrix} \approx \begin{pmatrix} 0 \\ 0 \\ \vdots \\ -\alpha_0 \end{pmatrix} = \vec{y}.$$

Observe that if $\|A\vec{x} - \vec{y}\|$ is small and the components $x_1, \dots, x_{n-1}$ are close to zero, this will consequently force $|L(\vec{x})|$ to be small as well. To establish a lower bound on the evaluations $|L(\vec{x})|$, it will be useful to understand how small $\|A\vec{x} - \vec{y}\|$ can become. However, we want to exclude the degenerate case in which the norm vanishes, i.e. $\|A\vec{x} - \vec{y}\| = 0$. Depending on whether we are in the homogeneous case, or the inhomogeneous one, degeneracies amount to the following:

- if $\alpha_0 = 0$, then $\vec{y} = \vec{0}$, thus $\|A\vec{x} - \vec{y}\| = 0 \iff \vec{x} = \vec{0}$. We will therefore require that the form L is not evaluated at the zero vector (assuming $\alpha_i \neq 0$ for all possible i).
- if we have an inhomogeneity term $\alpha_0 \neq 0$, then $\|A\vec{x} - \vec{y}\| = 0 \iff \alpha_0 + \alpha_n x_n = 0$. In the inhomogeneous case we thus require that the coefficients α_i, $i \in \{0, 1, \dots, n\}$ are linearly independent over $\mathbb{Z}$.

So let us assume that we are in a nondegenerate case. If $\vec{y}$ is a lattice vector, i.e. $\vec{y} \in \mathcal{L}(A)$, determining the smallest possible nonzero norm $\|A\vec{x} - \vec{y}\|$ is equivalent to solving the *shortest vector problem*. On the other hand, if $\vec{y} \notin \mathcal{L}(A)$, the problem shifts to one about *best approximation*—finding the distance from a vector outside $\mathcal{L}(A)$ to the lattice itself. At this stage we might be worried about the *best approximation* problem, since lattices are non-convex subsets of $\mathbb{R}^n$ and for that reason best approximations of vectors in $\mathbb{R}^n$ by elements of $\mathcal{L}(A)$ might not exist. This shift in perspective, however, still turns out to be advantageous, as lattice theory provides solutions for both scenarios. How to find a lower bound on $\|A\vec{x} - \vec{y}\|$ in these two cases will be the main topic of the next chapter.

In [4, Section V.3] it is pointed out that vectors $\vec{x}$ which make $|L(\vec{x})|$ small, will have $|x_n|$ very small, compared to the $|x_i|$ for $1 \leq i \leq n-1$. In such cases, Smart makes the suggestion to scale the last row of the above matrix A by a large factor $C \in \mathbb{N}$ and instead consider the matrix

$$A = \begin{pmatrix} 1 & & & 0 \\ & \ddots & & \\ 0 & & 1 & 0 \\ [C\alpha_1] & \dots & [C\alpha_{n-1}] & [C\alpha_n] \end{pmatrix} \in M_{n\times n}(\mathbb{Z}),$$

and vector

$$\vec{y} = (0, \dots, 0, -[C\alpha_0])^t \in \mathbb{Z}^n,$$

where $[a]$ stands for the nearest integer to a with any fixed convention on rounding up or down numbers of the form $z + 1/2, z \in \mathbb{Z}$. At this stage the choice of the constant C may appear arbitrary. In Section 3.2 we will see how the value of C is determined.

The above construction can be generalized also to complex linear forms

$$L(\vec{x}) = \alpha_0 + \alpha_1 x_1 + \cdots + \alpha_n x_n, \ \ \alpha_i \in \mathbb{C}.$$

If for some $\vec{x} \in \mathbb{Z}^n$, $|L(\vec{x})|$ is "small", then the real and imaginary parts of $L(\vec{x})$ should be "small" simultaneously. Approximation of both $\Re(L(\vec{x}))$ and $\Im(L(\vec{x}))$ will be captured by the matrix

$$A = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ 0 & \dots & 1 & 0 & 0 \\ [C\Re(\alpha_1)] & \dots & 0 & [C\Re(\alpha_{n-1})] & [C\Re(\alpha_n)] \\ [C\Im(\alpha_1)] & \dots & 0 & [C\Im(\alpha_{n-1})] & [C\Im(\alpha_n)] \end{pmatrix} \in M_{n\times n}(\mathbb{Z}).$$

Up to this point we have presented the concept of a lattice in $\mathbb{R}^n$ and motivated the setup for associating an approximation lattice to a linear form. Our discussion now segways to the applications of LLL-reduction, where in the next chapter the main results for approximating complex linear forms will be presented.

3 Applications of LLL-Reduction

This chapter is devoted to applications of LLL-reduction. Here we showcase the main results from lattice theory which will be used in Chapter 4. Main reference for this chapter is [4, Section V.3 and Section VI.3]

3.1 Approximation Using LLL

We start off our discussion with the following result:

Theorem 3.1. *Let $\mathcal{L}$ be a full rank lattice of dimension n and B be a matrix, the columns of which constitute a reduced basis for $\mathcal{L}$. Let B^* be the Gram-Schmidt matrix associated to B. Then*

1. *For $1 \leq j \leq i \leq n$ we have $\|\vec{b_j}\|^2 \leq 2^{i-1}\|b_i^*\|^2$.*
2. *$\Delta(\mathcal{L}) \leq \prod_{i=1}^{n} \|\vec{b_i}\| \leq 2^{n(n-1)/4}\Delta(\mathcal{L})$.*
3. *$\|\vec{b_1}\| \leq 2^{(n-1)/4}\Delta(\mathcal{L})^{1/n}$.*

Proof. We have shown in (2.1) that $\|b_i^*\|^2 \geq (\frac{3}{4} - \mu_{i,i-1}^2)\|\vec{b_{i-1}^*}\|^2$. By definintion of an LLL-reduced basis, $\mu_{i,i-1} \in [-\frac{1}{2}, \frac{1}{2}]$, hence $\frac{3}{4} - \mu_{i,i-1}^2 \geq \frac{1}{2}$ and one obtains

$$\|b_i^*\|^2 \geq (\frac{3}{4} - \mu_{i,i-1}^2)\|\vec{b_{i-1}^*}\|^2 \geq \frac{1}{2}\|\vec{b_{i-1}^*}\|^2.$$

Inductively, continuing the chain of inequalities

$$\|\vec{b_i^*}\|^2 \geq 2^{(i-1)-i}\|\vec{b_{i-1}^*}\|^2 \geq 2^{(i-2)-i}\|\vec{b_{i-2}^*}\|^2 \geq \cdots \geq 2^{j-i}\|\vec{b_j^*}\|^2 \quad \text{for} \quad 1 \leq j \leq i \leq n,$$

we conclude that

$$\|\vec{b_j^*}\|^2 \leq 2^{i-j}\|\vec{b_i^*}\|^2, \quad \text{for} \quad 1 \leq j \leq i \leq n. \tag{3.1}$$

From the definition of the Gram-Schmidt process, express

$$\vec{b_i} = \vec{b_i^*} + \sum_{j=1}^{i-1} \mu_{i,j}\vec{b_j^*},$$

take squared norms on both sides and use mutual orthogonality of the set $\{b_i^*\}_{i=1}^n$ to get, by the *Pythagorean identity*, that

$$\|\vec{b_i}\|^2 = \left\| \vec{b_i^*} + \sum_{j=1}^{i-1} \mu_{i,j}\vec{b_j^*} \right\|^2 = \|\vec{b_i^*}\|^2 + \sum_{j=1}^{i-1} |\mu_{i,j}|^2 \left\|\vec{b_j^*}\right\|^2. \tag{3.2}$$

Next, combine (3.1) and (3.2) with $|\mu_{i,j}|^2 \leq 2^{-2}$ to find that

$$\|\vec{b_i}\|^2 \leq \left(1 + \sum_{j=1}^{i-1} 2^{i-j-2}\right) \|\vec{b_i^*}\|^2 = \left(1 + \frac{1}{4}(2^i - 2)\right) \|\vec{b_i^*}\|^2. \tag{3.3}$$

Now, suppose for the sake for a contradiction that $1 + \frac{1}{4}(2^i - 2) = \frac{1}{2} + 2^{i-2} > 2^{i-1}$ to obtain that $\frac{1}{2^i} + \frac{1}{2} > 1$, which can not be for $i \geq 1$. With that in mind, (3.3) becomes $\|\vec{b_i}\|^2 \leq 2^{i-1}\|\vec{b_i^*}\|^2$. Ultimately, by combining the last inequality (with i replaced by j) with (3.1), in the order mentioned, one has

$$\|\vec{b_j}\|^2 \leq 2^{j-1}\|\vec{b_j^*}\|^2 \leq 2^{j-1+i-j}\|\vec{b_i^*}\|^2 \leq 2^{i-1}\|\vec{b_i^*}\|^2,$$

which is precisely what we had to prove in *1*.

After normalizing the vectors in B^*, we express B in terms of a QR-factorization $B = QR$, where Q is the matrix with columns - the normalized vectors $\dfrac{\vec{b_i^*}}{\|\vec{b_i^*}\|}$, and

$$R = \begin{pmatrix} \langle \vec{b_1}\ ,\ \dfrac{\vec{b_1^*}}{\|\vec{b_1^*}\|} \rangle & \langle \vec{b_2}\ ,\ \dfrac{\vec{b_1^*}}{\|\vec{b_1^*}\|} \rangle & \dots & \langle \vec{b_n}\ ,\ \dfrac{\vec{b_1^*}}{\|\vec{b_1^*}\|} \rangle \\ 0 & \langle \vec{b_2}\ ,\ \dfrac{\vec{b_2^*}}{\|\vec{b_2^*}\|} \rangle & \dots & \langle \vec{b_n}\ ,\ \dfrac{\vec{b_2^*}}{\|\vec{b_2^*}\|} \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \langle \vec{b_n}\ ,\ \dfrac{\vec{b_n^*}}{\|\vec{b_n^*}\|} \rangle \end{pmatrix}.$$

Note that $\langle \vec{b_i}\ ,\ \dfrac{\vec{b_i^*}}{\|\vec{b_i^*}\|} \rangle = \dfrac{1}{\|\vec{b_i^*}\|}\langle \vec{b_i}\ ,\ \vec{b_i^*} \rangle = \dfrac{1}{\|\vec{b_i^*}\|}\langle \vec{b_i^*} + \sum\limits_{j=1}^{i-1} \mu_{i,j}\vec{b_j^*}\ ,\ \vec{b_i^*} \rangle = \dfrac{\langle \vec{b_i^*}\ ,\ \vec{b_i^*} \rangle}{\|\vec{b_i^*}\|} = \|\vec{b_i^*}\|$. Thus, the diagonal entries of the upper-triangular matrix R are $\|\vec{b_i^*}\|$, from which follows that

$$\det(R) = \prod_{i=1}^{n} \|\vec{b_i^*}\|.$$

Therefore, $\det(B) = \det(Q)\det(R)$, with Q-orthogonal, so $\det(Q) = \pm 1$, and we conclude that $\Delta\mathcal{L}(B) = |\det(B)| = |\det(R)| = \prod\limits_{i=1}^{n} \|\vec{b_i^*}\|$. Now, it is clear form the definition of the Gram-Schmidt process that $\|\vec{b_i^*}\| \leq \|\vec{b_i}\|$, so combining this inequality with the result in *1*, we obtain

$$\Delta\mathcal{L}(B) \leq \prod_{i=1}^{n} \|\vec{b_i}\| \leq \prod_{i=1}^{n} 2^{(i-1)/2}\|\vec{b_i^*}\| = 2^{\sum_{i=1}^{n} \frac{i-1}{2}} \cdot \prod_{i=1}^{n} \|\vec{b_i^*}\| = 2^{n(n-1)/4} \cdot \Delta\mathcal{L}(B).$$

Ultimately, $\Delta\mathcal{L}(B) \leq \prod\limits_{i=1}^{n} \|\vec{b_i}\| \leq 2^{n(n-1)/4} \cdot \Delta\mathcal{L}(B)$, completing the proof of *2*.

To finalize the proof of the Theorem, take statement *1.* with $j = 1$, so that

$$\|\vec{b_1}\|^2 \leq 2^{i-1}\|\vec{b_i^*}\|^2,$$

and take the product over all $i \in \{1, 2, \ldots, n\}$ to conclude that

$$\|\vec{b_1}\|^{2n} \leq \prod_{i=1}^{n} 2^{i-1}\|\vec{b_i^*}\|^2 = 2^{\sum_{i=1}^{n}(i-1)} \cdot \prod_{i=1}^{n} \|\vec{b_i}^*\|^2 = 2^{\frac{n(n-1)}{2}} \Delta\mathcal{L}(B)^2,$$

from which follows the statement in *3.* □

For now introducing Theorem 3.1 seems unmotivated. We will see shortly (Section 3.2) that with the help of result *3.* in the above theorem we will be able to determine the value of the constant C from Section 2.2.

Next, we shift our attention to the problem of approximating the first successive minimum M_1. The next result tells us that the first vector inside an LLL-reduced lattice basis provides a good estimate for a nonzero lattice vector achieving minimal norm.

Theorem 3.2. *Let $\mathcal{L}$ be a full rank lattice of dimension n and B be a matrix, the columns of which constitute a reduced basis for $\mathcal{L}$. Let B^* be the Gram-Schmidt matrix associated to B. Define the constant*

$$c_1 \stackrel{\text{def}}{=\!=} \max\left\{ \frac{\|\vec{b_1}\|^2}{\|\vec{b_i^*}\|^2} : 1 \leq i \leq n \right\}.$$

Then $\forall \vec{x} \neq \vec{0},\ \vec{x} \in \mathcal{L}$, we have the inequality $\|\vec{b_1}\|^2 \leq c_1\|\vec{x}\|^2$.

Proof. By definition of c_1 we have that $c_1 \geq \dfrac{\|\vec{b_1}\|^2}{\|\vec{b_i^*}\|^2}$ for all values of i, so $\|\vec{b_1}\|^2 \leq c_1\|\vec{b_i^*}\|^2, \forall i \in \{1, 2, \ldots, n\}$. Since the collection $\{\vec{b_i}\}_{i=1}^n$ is a basis for the lattice $\mathcal{L}(B)$ and $\{\vec{b_i^*}\}_{i=1}^n$ is a basis for $\mathbb{R}^n$, we can write for any $\vec{x} \in \mathcal{L}(B)$:

$$\vec{x} = \sum_{i=1}^{n} r_i\vec{b_i} = \sum_{i=1}^{n} r_i'\vec{b_i^*}, \tag{3.4}$$

where $r_i \in \mathbb{Z}$ and $r_i' \in \mathbb{R}$. Let i_0 denote the largest index such that $r_i \neq 0$. From the Gram-Schmidt process we can express each $\vec{b_i}$ in terms of the vectors $\vec{b_i^*}$ as

$$\vec{b_i} = \vec{b_i^*} + \sum_{j=1}^{i-1} \mu_{i,j}\vec{b_j^*}.$$

With that in mind, one has the following expression for $\vec{x}$:

$$\begin{aligned}\vec{x} = \vec{b_1^*}\left[r_1 + r_2\mu_{2,1} + ... + r_n\mu_{n,1}\right] + \\ + \vec{b_2^*}\left[r_2 + r_3\mu_{3,2} + ... + r_n\mu_{n,2}\right] + \\ \vdots \\ + \vec{b_{n-1}^*}\left[r_{n-1} + r_n\mu_{n,n-1}\right] + \vec{b_n^*}r_n = \sum_{i=1}^{n} r_i'\vec{b_i^*}.\end{aligned}$$

Uniqueness of the basis representation in B^* and coefficient comparison with (3.4) give rise to the system

$$\begin{cases} r_1' = r_1 + r_2\mu_{2,1} + ... + r_n\mu_{n,1} \\ r_2' = r_2 + r_3\mu_{3,2} + ... + r_n\mu_{n,2} \\ \vdots \\ r_{n-1}' = r_{n-1} + r_n\mu_{n,n-1} \\ r_n' = r_n \end{cases}.$$

From the above system we see that if $i_0 = n$, then $r_{i_0} = r_n = r_n' = r_{i_0}'$. On the other hand, $i_0 = n-1$ forces $r_n = r_n' = 0$ and consequently $r_{i_0} = r_{n-1} = r_{n-1}' = r_{i_0}'$ form the second to last equation in the system above. Continuing inductively in this manner, we conclude that for any possible value of $i_0 \in \{1, 2, \ldots, n\}$ one has $r_{i_0} = r_{i_0}'$. Then, from the *Pythagorean identity* (using the mutual orthogonality of the vectors $\vec{b_i^*}$)

$$\|\vec{x}\|^2 = \left\|\sum_{i=1}^{n} r_i'\vec{b_i^*}\right\|^2 = \sum_{i=1}^{n}(r_i')^2\|\vec{b_i^*}\|^2 \geq (r_{i_o}')^2\|\vec{b_{i_0}^*}\|^2 = (r_{i_o})^2\|\vec{b_{i_0}^*}\|^2 \geq \|\vec{b_{i_0}^*}\|^2 \geq c_1^{-1}\|\vec{b_1}\|^2.$$

Ultimately, $\|\vec{b_1}\|^2 \leq c_1\|\vec{x}\|^2$, which completes the proof. □

In view of Theorem 3.2, for $\vec{x}$ - a nonzero lattice vector, one has $\|\vec{b_1}\| \leq \sqrt{c_1}\|\vec{x}\|$. When we perform calculations we will have the value of the constant c_1 very close to 1. This in turn will tell us that the first vector in a reduced lattice basis can be seen as an approximation of a nonzero lattice vector, the norm of which achieves the first successive minimum.

In Section 2.2 we mentioned the problem with *best approximation* of a vector in $\mathbb{R}^n$ with an element of a lattice $\mathcal{L} \subset \mathbb{R}^n$. In particular, non-convexity of the lattice, as a subset of $\mathbb{R}^n$, tells us that best approximations may not exist. Fortunately enough, the next theorem will give us an estimate for the distance from a vector, which does not lie inside a lattice, to the lattice itself. This result will come in use in Section 3.2 when we approximate inhomogeneous complex linear forms.

Theorem 3.3. *Let $\mathcal{L}$ be a full rank lattice of dimension n and B - a reduced basis for $\mathcal{L}$. Let B^* be the Gram-Schmidt basis associated to B. Assume a vector $\vec{y} \notin \mathcal{L}$ is given*

and define the vector $\vec{\sigma} \stackrel{\text{def}}{=\!=} B^{-1}\vec{y}$. *Let* i_0 *be the largest index for which* $\{\vec{\sigma_{i_0}}\} \neq 0$, *where* $\{r\}$ *denotes the distance from the number* $r \in \mathbb{R}$ *to its nearest integer. Then* $\forall \vec{x} \in \mathcal{L}$ *one has the inequality*

$$\|\vec{x} - \vec{y}\|^2 \geq c_1^{-1}\{\vec{\sigma}_{i_0}\}\|\vec{b_1}\|^2.$$

Proof. Let $\vec{x} \in \mathcal{L}$ be arbitrary and express it relative to the bases B (a basis of $\mathcal{L}$) and B^* (a basis of $\mathbb{R}^n$) in two different ways

$$\vec{x} = \sum_{i=1}^{n} r_i\vec{b_i} = \sum_{i=1}^{n} r_i'\vec{b_i^*},$$

with $r_i \in \mathbb{Z}$ and $r_i' \in \mathbb{R}$. By construction, $\vec{\sigma} = B^{-1}\vec{y}$ and we can also represent $\vec{y}$ relative to the bases B, and B^* accordingly

$$\vec{\sigma} = B^{-1}\vec{y} \iff \vec{y} = B\vec{\sigma} = \sum_{i=1}^{n} \sigma_i\vec{b_i} = \sum_{i=1}^{n} \sigma_i'\vec{b_i^*},$$

the latter expression defining the constants $\sigma_i' \in \mathbb{R}$. Note that the assumption on $\vec{y} \notin \mathcal{L}(B)$ implies that the vector $\vec{\sigma}$ has at least one non-integer entry. Call i_1 the largest index for which $r_i \neq \sigma_i$. We will now show that $r_{i_1} - \sigma_{i_1} = r_{i_1}' - \sigma_{i_1}'$. In the expression $\vec{x} - \vec{y} = \sum_{i=1}^{n}(r_i - \sigma_i)\vec{b_i}$, we can express each $\vec{b_i}$ in terms of the vectors $\vec{b_i}^*$ from the Gram-Schmidt process. After collecting terms we obtain a representation of $\vec{x} - \vec{y}$ in terms of the $\mathbb{R}^n$-basis B^* as follows:

$$\begin{aligned}
\vec{x} - \vec{y} = {} & \vec{b_1^*}\left[r_1 - \sigma_1 + (r_2 - \sigma_2)\,\mu_{2,1} + \ldots + (r_n - \sigma_n)\,\mu_{n,1}\right] + \\
& + \vec{b_2^*}\left[r_2 - \sigma_2 + (r_3 - \sigma_3)\,\mu_{3,2} + \ldots + (r_n - \sigma_n)\,\mu_{n,2}\right] + \\
& \vdots \\
& + \vec{b_{n-1}^*}\left[r_{n-1} - \sigma_{n-1} + (r_n - \sigma_n)\,\mu_{n,n-1}\right] + \\
& + \vec{b_n^*}\left[r_n - \sigma_n\right] = \sum_{i=1}^{n}(r_i' - \sigma_i')\,\vec{b_i^*},
\end{aligned}$$

with the last equality coming from the initial representations of $\vec{x}$ and $\vec{y}$ in terms of the columns of B^*. Uniqueness of basis representation (in $\mathbb{R}^n$) and coefficient comparison give rise to the following system:

$$\begin{cases}
r_1' - \sigma_1' = r_1 - \sigma_1 + (r_2 - \sigma_2)\,\mu_{2,1} + \ldots + (r_n - \sigma_n)\,\mu_{n,1} \\
r_2' - \sigma_2' = r_2 - \sigma_2 + (r_3 - \sigma_3)\,\mu_{3,2} + \ldots + (r_n - \sigma_n)\,\mu_{n,2} \\
\vdots \\
r_{n-1}' - \sigma_{n-1}' = r_{n-1} - \sigma_{n-1} + (r_n - \sigma_n)\,\mu_{n,n-1} \\
r_n' - \sigma_n' = r_n - \sigma_n
\end{cases}.$$

By definition, i_1 is the largest index i, such that $r_i \neq \sigma_i$. If $i_1 = n$, then $r_{i_1} - \sigma_{i_1} = r_{i_1}' - \sigma_{i_1}'$, evidently. If $i_1 = n-1$, then $r_n - \sigma_n = 0$ forces $r_n' - \sigma_n' = 0$ and the second to last equation

in the above system tells us that $r'_{n-1} - \sigma'_{n-1} = r_{n-1} - \sigma_{n-1}$. Repeatedly arguing in this manner we descend to any $i_1 \in \{1, 2, ..., n\}$ with the conclusion that $r_{i_1} - \sigma_{i_1} = r'_{i_1} - \sigma'_{i_1}$, which is the claim we wanted to prove.

Now, from the *Pythagorean identity*,

$$\|\vec{x} - \vec{y}\|^2 = \left\| \sum_{i=1}^{n} (r'_i - \sigma'_i)\vec{b_i^*} \right\|^2 = \sum_{i=1}^{n} |r'_i - \sigma'_i|^2 \|\vec{b_i^*}\|^2 \geq |r'_{i_1} - \sigma'_{i_1}|^2 \|\vec{b_{i_1}^*}\|^2 =$$
$$= |r_{i_1} - \sigma_{i_1}|^2 \|\vec{b_{i_1}^*}\|^2 \geq |r_{i_1} - \sigma_{i_1}|^2 \ \cdot c_1^{-1} \cdot \|\vec{b_1}\|^2.$$

The last inequality follows from the definition of the constant c_1. Suppose now that $i_1 < i_0$. In this case, $\sigma_{i_0} = r_{i_0} \in \mathbb{Z}$. At the same time, i_0 was chosen such that $\{\sigma_{i_0}\} \neq 0$, i.e., $\sigma_{i_0} \notin \mathbb{Z}$, which contradicts the line above. In the case of $i_1 = i_0$, we have $|r_{i_1} - \sigma_{i_1}| = |r_{i_0} - \sigma_{i_0}| \geq \{\sigma_{i_0}\}$, since $r_{i_0} \in \mathbb{Z}$ and $\{\sigma_{i_0}\}$ denotes the distance from σ_{i_0} to the nearest integer. Thus, in this case

$$\|\vec{x} - \vec{y}\|^2 \geq c_1^{-1} \{\sigma_{i_0}\} \|\vec{b_1}\|^2, \tag{3.5}$$

which is the claim of the theorem we are proving. Lastly, if $i_1 > i_0$ we have that $\{\sigma_{i_1}\} = 0$, so $\sigma_{i_1} \in \mathbb{Z}$ and $\sigma_{i_1} - r_{i_1} \neq 0$, together with $r_{i_1} \in \mathbb{Z}$, implies that $|\sigma_{i_1} - r_{i_1}| > 1 > \frac{1}{2} \geq \{\sigma_{i_0}\}$, which again gives inequality (3.5). □

At this stage we have approximated a nonzero lattice vector of minimal norm. Then we gave lower bounds for the first successive minimum and the distance from vector to lattice in terms of the constant c_1 and the norm of the first vector in an LLL - reduced basis. With the help of the last three theorems, the discussion continues to our next application of LLL - reduction, namely approximating inhomogeneous complex linear forms.

3.2 Approximating Complex Linear Forms

Let us start with an inhomogeneous complex linear form

$$L(\vec{x}) = \alpha_0 + \sum_{i=1}^{n} \alpha_i x_i, \quad \text{with} \quad \alpha_i \in \mathbb{C} \ \forall i \in \{1, \ldots, n\}, \quad \alpha_0 \neq 0 \quad \text{and} \quad n \geq 2, \tag{3.6}$$

evaluated at a vector $\vec{x} \in \mathbb{Z}^n$, the components of which are bounded by some positive constants $x_i \leq X_i$, for $i = 1, \ldots, n$. Define $X_0 \stackrel{\text{def}}{=\!=} \max_{i=1}^{n} \{X_i\}$. As we pointed out in Section 2.2, in Chapter 4 we will be interested in determining an upper bound on $A \in \mathbb{N}$ from an inequality of the form

$$|L\vec{x}| \leq c_2 e^{-c_3 A},$$

for some positive constants c_2 and c_3, held fixed. In the current section we will give an answer to this problem for the more general case where A is replaced by a power H^q, for $q \in \mathbb{N}$. We then determine an upper bound on H.

We start by a choice of $C \approx X_0^{n/2}$ ("$\approx$" to stand for "approximately equal to"). Then we associate to the form L from (3.6) the approximation lattice $\mathcal{L}(A)$, generated by the matrix

$$A = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ 0 & \dots & 1 & 0 & 0 \\ [C\Re(\alpha_1)] & \dots & 0 & [C\Re(\alpha_{n-1})] & [C\Re(\alpha_n)] \\ [C\Im(\alpha_1)] & \dots & 0 & [C\Im(\alpha_{n-1})] & [C\Im(\alpha_n)] \end{pmatrix} \in M_{n\times n}(\mathbb{Z}).$$

If necessary, we should relabel the coefficients α_i to make the determinant

$$\det(A) = [C\Re(\alpha_{n-1})][C\Im(\alpha_n)] - [C\Re(\alpha_n)][C\Im(\alpha_{n-1})] \neq 0.$$

This will guarantee linear independence for the columns of A, making $\mathcal{L}(A)$ a bona fide lattice of rank n. Then, compute a reduced basis $\{\vec{b_i}\}_{i=1}^n$ for the lattice $\mathcal{L}(A)$ and its associated Gram-Schmidt basis $\{\vec{b_i^*}\}_{i=1}^n$. Notice that the choice of C makes

$$\Delta(\mathcal{L}) = |\det(A)| = \left|[C\Re(\alpha_{n-1})][C\Im(\alpha_n)] - [C\Re(\alpha_n)][C\Im(\alpha_{n-1})]\right| \leq K_1 \cdot X_0^n,$$

for some positive constant $K_1 \in \mathbb{R}$. Referring to part *3.* of Theorem (3.1) we see that $\|\vec{b_1}\| \leq 2^{(n-1)/4}\Delta(\mathcal{L})^{1/n} \leq 2^{(n-1)/4}(K_1 \cdot X_0^n)^{1/n} = K_2 \cdot X_0$, for the constant $K_2 \stackrel{\text{def}}{=\!=} 2^{(n-1)/4}K_1^{1/n}$. This shows us that the norm of the first vector in the reduced basis is bounded above by a constant multiple of X_0.

Finally, define the vector $\vec{y} = (0, \dots, 0, -[C\Re(\alpha_0)], -[C\Im(\alpha_0)])^t$. Let $\vec{x} \in \mathcal{L}(A)$ be arbitrary. If $\vec{y} \in \mathcal{L}(A)\backslash\{\vec{x}\}$, Theorem 3.2 gives us a constant c_4, such that $\|\vec{x} - \vec{y}\| \geq c_4$. Alternatively, if $\vec{y} \notin \mathcal{L}(A)$ the constant c_4 for which $\|\vec{x} - \vec{y}\| \geq c_4$ comes from Theorem 3.3. With the above definitions we have all the tools necessary to prove the next result.

Lemma 3.4. *Let $L(\vec{x})$ be the form given in* (3.6) *and suppose that $|L(\vec{x})| \leq c_2 e^{-c_3 H^q}$ for fixed positive constants c_2, c_3, $q \in \mathbb{N}$. Let $S = \sum_{i=1}^{n-2} X_i^2$ and $T = (1 + \sum_{i=1}^n X_i)/\sqrt{2}$. If $c_4^2 \geq T^2 + S$ then either*

$$H \leq \sqrt[q]{\frac{1}{c_3}\left(\log(Cc_2) - \log\left(\sqrt{c_4^2 - S} - T\right)\right)}, \quad \textit{or} \quad x_1 = \dots = x_{n-2} = 0, \quad \textit{and}$$

$$x_{n-1}[C\Re(\alpha_{n-1})] + x_n[C\Re(\alpha_n)] = [C\Re(\alpha_0)],$$
$$x_{n-1}[C\Im(\alpha_{n-1})] + x_n[C\Im(\alpha_n)] = [C\Im(\alpha_0)].$$

Proof. Define $\Phi_1 = [C\Re(\alpha_0)] + \sum_{i=1}^n x_i[C\Re(\alpha_i)]$ and $\Phi_2 = [C\Im(\alpha_0)] + \sum_{i=1}^n x_i[C\Im(\alpha_i)]$. To simplify notation, define $B_1(\alpha_i) \stackrel{\text{def}}{=\!=} [C\Re(\alpha_i)] - C\Re(\alpha_i)$ and $B_2(\alpha_i) \stackrel{\text{def}}{=\!=} [C\Im(\alpha_i)] - C\Im(\alpha_i), \forall i \in \{0, 1, \dots, n\}$. It is immediate that $|B_1(\alpha_i)|, |B_2(\alpha_i)| \leq 1/2$ for all i. Apply the triangle inequality to get

$$\left|\Phi_1 + i\Phi_2 - C\left(\alpha_0 + \sum_{i=1}^n x_i\alpha_i\right)\right| = \left|B_1(\alpha_0) + iB_2(\alpha_0) + \sum_{i=1}^n x_i\left(B_1(\alpha_i) + iB_2(\alpha_i)\right)\right| \leq$$

$$\leq \sqrt{B_1(\alpha_0)^2 + B_2(\alpha_0)^2} + \sum_{i=1}^n x_i\sqrt{B_1(\alpha_i)^2 + B_2(\alpha_i)^2} \leq \frac{1}{\sqrt{2}} + \sum_{i=1}^n X_i \cdot \frac{1}{\sqrt{2}} = T,$$

where in the last step we used that both $|B_1(\alpha_i)|, |B_2(\alpha_i)| \leq 1/2$ and $|x_i| \leq X_i, \forall i$. Now, by the reverse triangle inequality,

$$|\Phi_i+i\Phi_2|-\left|C(\alpha_0+\sum_{i=1}^{n} x_i\alpha_i)\right| \leq \left||\Phi_i+i\Phi_2|-|C(\alpha_0+\sum_{i=1}^{n} x_i\alpha_i)|\right| \leq |\Phi_i+i\Phi_2-C(\alpha_0+\sum_{i=1}^{n} x_i\alpha_i)| \leq T,$$

and rearranging terms gives rise to the following inequality

$$|\Phi_1 + i\Phi_2| \leq T + C|\alpha_0 + \sum_{n}^{i=1} x_i\alpha_i| \leq T + C \cdot c_2 \cdot e^{-c_3 H^q}. \tag{3.7}$$

Consider further the lattice point $\vec{x} = A\vec{z}$, where $\vec{z} = (x_1, \ldots, x_n)^t$. It follows that

$$\vec{x} = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ 0 & \ldots & 1 & 0 & 0 \\ [C\Re(\alpha_1)] & \ldots & 0 & [C\Re(\alpha_{n-1})] & [C\Re(\alpha_n)] \\ [C\Im(\alpha_1)] & \ldots & 0 & [C\Im(\alpha_{n-1})] & [C\Im(\alpha_n)] \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_{n-2} \\ [C\Re(\alpha_1)]x_1 + \ldots [C\Re(\alpha_n)]x_n \\ [C\Im(\alpha_1)]x_1 + \ldots [C\Im(\alpha_n)]x_n \end{pmatrix}$$

and

$$\vec{x} - \vec{y} = (x_1, \ldots, x_{n-2}, \Phi_1, \Phi_2)^t.$$

Now, either $\vec{x} = \vec{y}$, forcing $x_1 = \cdots = x_{n-2} = 0$ and

$$x_{n-1}[C\Re(\alpha_{n-1})] + x_n[C\Re(\alpha_n)] = [C\Re(\alpha_0)],$$
$$x_{n-1}[C\Im(\alpha_{n-1})] + x_n[C\Im(\alpha_n)] = [C\Im(\alpha_0)],$$

or $\vec{x} - \vec{y} \neq \vec{0}$, in which case, upon expanding the Euclidean norm in $\mathbb{R}^n$,

$$c_4^2 \leq \|\vec{x} - \vec{y}\|^2 = \sum_{i=1}^{n-2} x_i^2 + \Phi_1^2 + \Phi_2^2 \leq \sum_{i=1}^{n-2} X_i^2 + \Phi_1^2 + \Phi_2^2 = S + |\Phi_1 + i\Phi_2|^2.$$

Combine the above line with (3.7) to get $c_4^2 \leq S + \left(T + C \cdot c_2 \cdot e^{-c_3 H^q}\right)^2$. Now, if $c_4^2 - S \geq T^2 \geq 0$, positivity of $T + C \cdot c_2 \cdot e^{-c_3 H^q}$ allows us to conclude that

$$\sqrt{c_4^2 - S} \leq T + C \cdot c_2 \cdot e^{-c_3 H^q} \iff e^{-c_3 H^q} \geq \frac{1}{Cc_2}\left(\sqrt{c_4^2 - S} - T\right),$$

in which, after taking the natural logarithm on both sides, one concludes that

$$H \leq \sqrt[q]{\frac{1}{c_3}\left(\log(Cc_2) - \log\left(\sqrt{c_4^2 - S} - T\right)\right)}$$

and completes the proof. □

In this section we obtained the main results for approximating inhomogeneous complex linear forms. Next, in Chapter 4, we will see how these results can be applied in the context of Diophantine approximation.

4 The Method of Tzanakis and de Weger

Nihil est in arithmetico quod non prius fuerit in algebraico.

André Weil [12]

The aim of this chapter is to present the method of Tzanakis and de Weger for solving Thue equations and apply it on concrete examples. Section 4.1 is devoted to derivation of the method. In Section 4.2 we apply this method to solve the Diophantine equation $4X^4 - 3Y^4 = \pm 61$. We close this chapter with a short analysis on how certain choices can affect the performance of the method and make a comment on our `Mathematica` implementation. Main reference is Section VII.1 of the book by Nigel Smart [4]. Complementary literature includes [8], [15], [17] and [19].

4.1 The Theory

To set the scene, consider a Diophantine equation in the unknowns X and Y, of the form

$$F(X, Y) = m, \tag{4.1}$$

where $F(X, Y) \in \mathbb{Z}[X, Y]$ is a binary form having at least 3 distinct complex roots and $m \in \mathbb{Z}$ is a nonzero integer. We shall assume that $F(X, Y)$ is irreducible in $\mathbb{Q}[X, Y]$, where the case of F - reducible would follow from considering each irreducible component separately, with some extra care for the case in which F factors as a product of two binary forms of degree 2. With this further irreducibility condition, equation (4.1) is referred to as a Thue equation. We can always reduce the problem to the case in which $F(X, 1) \in \mathbb{Z}[X]$ is monic, via the following re-scaling argument: let

$$F(X, Y) = a_0 X^n + a_{n-1} X^{n-1} Y + \cdots + a_n Y^n = m$$

and multiply both sides of the equation by a_0^{n-1} to obtain

$$a_0^n X^n + a_{n-1}(a_0 X)^{n-1} Y + \cdots + a_n a_0^{n-1} Y^n = m a_0^{n-1}.$$

Now, introducing the new variable $X' = a_0 X$, the initial equation (4.1) reduces to solving the Diophantine equation

$$(X')^n + a_{n-1}(X')^{n-1} Y + \cdots + a_{n-1} a_0^{n-2} (X') Y^{n-1} + a_n a_0^{n-1} Y^n = m a_0^{n-1}$$

and rescaling its solutions $(X', Y) \in \mathbb{Z}^2$ will give us all solutions $(X'/a_0, Y) \in \mathbb{Z}^2$ for the Diophantine equation (4.1). From this point onwards, we assume $F(X, 1) \in \mathbb{Z}[X]$ to be monic.

Let us now introduce the theorem of Axel Thue.

Theorem 4.1 (Thue)**.** *If $F(X, Y)$ is a homogeneous polynomial with integer coefficients, irreducible over the rationals and of degree > 2 and m is a non-zero integer, then $F(X, Y) = m$ has only a finite number of integer solutions.*

A proof of Thue's Theorem can be fond in [19, Chapter 4]. The rest of this section is devoted to the derivation of the method of de Weger and Tzanakis for finding all solutions to a given Thue equation.

Until the end of this chapter we set θ to be any of the roots of $F(X, 1)$ and let $\theta^{(1)}, \ldots, \theta^{(n)}$ denote its Galois conjugates. Assuming that $F(X, Y)$ is irreducible over $\mathbb{Q}[X, Y]$ forces $F[X, 1]$ - irreducible over $\mathbb{Q}[X]$, since, if not, say $F(X, 1) = A(X)B(X)$ is a factorization in $\mathbb{Q}[X]$, then $F(X, Y) = Y^n A(\frac{X}{Y})B(\frac{X}{Y})$ gives both a factorization in $\mathbb{Q}[X, Y]$ and a contradiction. Therefore $F(X, 1)$ is irreducible over $\mathbb{Q}$ and so it is the minimal polynomial of the algebraic integer θ, using the convention for minimal polynomials being monic with coefficients in $\mathbb{Q}$. Consequently $\theta^{(1)}, \ldots, \theta^{(n)}$ are the n roots of $F(X, 1)$. Further, let $F(X, 1)$ have s real and t pairs of complex conjugate non-real roots (i.e. $n = s + 2t$). It is widely accepted to order the conjugates of θ in the following way:

$$\theta^{(i)} \in \mathbb{R} \quad \text{for} \quad 1 \leq i \leq s,$$
$$\theta^{(i)} = \overline{\theta^{(i+t)}} \in \mathbb{C}\backslash\mathbb{R} \quad \text{for} \quad s + 1 \leq i \leq s + t.$$

One particularly easy case occurs when the polynomial $F(X, 1)$ has no real roots. This is illustrated with the following lemma:

Lemma 4.2. *Suppose that the polynomial $F(X, 1)$ has no real roots. Then, if $(X, Y) \in \mathbb{Z}^2$ solves the equation $F(X, Y) = m$, one has the inequality*

$$|Y| \leq \frac{|m|}{\min\limits_{i=1}^{n} |\Im(\theta^{(i)})|}.$$

Proof. Suppose $F(X, 1)$ has no real roots. We can factorise $F(X, 1)$ over $\mathbb{C}$ as

$$F(X, 1) = \prod_{i=1}^{n} \left(X - \theta^{(i)}\right)$$

and then, since $F(X, Y)$ is homogeneous, we have that $F(X, Y) = Y^n F(\frac{X}{Y}, 1)$, and so

$$F(X, Y) = Y^n \prod_{i=1}^{n} \left(\frac{X}{Y} - \theta^{(i)}\right) = \prod_{i=1}^{n} \left(X - \theta^{(i)} Y\right).$$

Let us have a solution $(X, Y) \in \mathbb{Z}^2$ to $F(X, Y) = m$. If $|X - \theta^{(i)} Y| > |m|$ for all $1 \leq i \leq n$, then the equality $F(X, Y) = m$ cannot hold. Therefore we have $|X - \theta^{(j)} Y| \leq |m|$ for at

least one $j \in \{1, \ldots, n\}$. For the same integers X, Y and any $\alpha + \beta i \in \mathbb{C}$, with $\alpha, \beta \in \mathbb{R}$, one has

$$|X - (\alpha + \beta i)Y| = |(X - \alpha Y) + (\beta Y)i| = \sqrt{(X - \alpha Y)^2 + (\beta Y)^2} \geq \sqrt{(\beta Y)^2} = |\beta Y|$$

and in particular, replacing $\alpha + \beta i$ with $\theta^{(j)}$ (which has nonzero imaginary part by hypothesis on the roots), we get

$$|\Im(\theta^{(j)})Y| \leq |X - \theta^{(j)}Y| \leq |m|, \quad \text{thus} \quad |Y| \leq \frac{|m|}{|\Im(\theta^{(j)})|} \leq \frac{|m|}{\min\limits_{i=1}^{n} |\Im(\theta^{(i)})|},$$

with $\min\limits_{i=1}^{n} |\Im(\theta^{(i)})| \neq 0$ by hypothesis on the roots of $F(X, 1)$. □

In view of Lemma 4.2 we see that if $F(X, 1)$ has no real roots, then we can bound the possible values for $Y \in \mathbb{Z}$ and iterating over those values of Y we obtain algebraic equations for X, to which we can easily find all possible integral solutions. Therefore, we will make one further assumption on the polynomial $F(X, 1)$ in requiring that it has at least one real root, i.e., $s \geq 1$.

Define the number field $K = \mathbb{Q}(\theta)$ and let $\sigma_1, \ldots, \sigma_n$ denote the n field embeddings from K into $\mathbb{C}$. Further, denote by $K^{(i)} = \sigma_i(K)$ the conjugate fields of K. Until the end of this chapter we shall assume that the pair $(X, Y) \in \mathbb{Z}^2$ solves (4.1). Define $\beta = X - \theta Y$ to get

$$\beta^{(i)} \stackrel{\text{def}}{=\!=} \sigma_i(\beta) = \sigma_i(X - \theta Y) = X - \theta^{(i)}Y$$

for the n conjugates of β, using the fact that the field embeddings act as identity on $\mathbb{Q}$. Since (X, Y) solves $F(X, Y) = m$, we see that $\beta^{(i)}$ has norm m:

$$N_{K/\mathbb{Q}}\left(\beta^{(i)}\right) = N\left(\beta^{(i)}\right) \stackrel{\text{def}}{=\!=} \prod_{j=1}^{n} \beta^{(j)} = \prod_{j=1}^{n} \left(X - \theta^{(j)}Y\right) = F(X, Y) = m.$$

Next we show that there are only finitely many nonassociate elements in O_K of norm m. For that we prove the following:

Proposition 4.2.1. *Let K be a number field and O_K its ring of algebraic integers. Then there are only finitely many nonzero ideals in O_K of norm $m \in \mathbb{N}$.*

Proof. Let $I \subset O_K$ be an ideal of norm $m \in \mathbb{N}$. Equivalently, we have that the cardinality $|O_K/I| = m$. Therefore, as an additive Abelian group, since $0 + I$ is the (additive) identity of O_K/I, we have that

$$ord\,(O_K/I) \cdot (1 + I) = m \cdot (1 + I) = m + I = 0 + I$$

and in particular, $\langle m \rangle \subset I$. But rings of algebraic integers of number fields are Dedekind domains, so "to contain is to divide" and we conclude that $I \,|\, \langle m \rangle$. Then factorise $\langle m \rangle$ into a product of prime ideals $\langle m \rangle = P_1 ... P_{k_1}$ and, by unique factorization of ideals in O_K, $I \,|\, \langle m \rangle$ tells us that there are only finitely many possibilities for the ideal I. In conclusion, there are only finitely many nonzero ideals in O_K of norm m. □

Now, back to our initial setting, let $\alpha \in O_K$ be an element of norm $m \in \mathbb{Z}\backslash\{0\}$. Then $|m| = |N(\alpha)| = N(\langle\alpha\rangle)$. According to Proposition 4.2.1, there are only finitely many ideals in O_K of norm $|m|$, and we may call them $Q_1, \ldots, Q_{k_2}$. Suppose now that α and β are two elements of norm m. Then both principal ideals $\langle\alpha\rangle$ and $\langle\beta\rangle$ have norm $|m|$ and so must be one of the Q_i's. If $Q_i = \langle\alpha\rangle \neq \langle\beta\rangle = Q_j$, then α and β are nonassociate elements. On the contrary, if $Q_i = \langle\alpha\rangle = \langle\beta\rangle$, then α and β are associates. Since there are only finitely many Q_i's, by assuming that there are infinitely many nonassociate elements in O_K of norm $\pm m$, we get infinitely many distinct principal ideals of norm $|m|$ and a contradiction. Hence, we conclude that there are only finitely many nonassociate elements $\mu_1, \ldots, \mu_{k_3}$ of O_K, such that $N(\mu_i) = \pm m$. Those μ_i of norm $+m$ constitute a complete set of nonassociate elements of O_K of norm m.

So we see that, since β has norm m, it can be expressed in the form $\beta = \mu\epsilon$, with $\epsilon \in O_K^*$ and μ coming from a complete set of nonassociate elements in O_K of norm m, which is finite. At this stage we recall *Dirichlet's Unit Theorem*:

Theorem 4.3 (Dirichlet)**.** *Let K be an algebraic number field of degree n. Let s be the number of real conjugate fields of K and $2t$ the number of complex non-real conjugate fields of K. Then O_K contains $s + t - 1$ units $\eta_1, \ldots, \eta_{s+t-1}$ such that each unit of O_K can be expressed uniquely in the form $\xi\eta_1^{a_1} \ldots \eta_{s+t-1}^{a_{s+t-1}}$, where ξ is a root of unity of O_K (a unit of finite order) and $a_1, \ldots, a_{s+t-1}$ are integers.*

We refer to the set of generators for the free part of O_K^* in Dirichlet's Theorem as *fundamental units*. For a proof of the theorem using properties of valuations, we refer to [8, Chapter 13]. Using an algebra computer package, such as `SageMath` or `Mathematica`, we are able to compute a complete set of representatives $\{\mu_i\}$ of the elements in O_K of norm m, up to associates, a set of fundamental units $\{\eta_i\}_{i=1}^{s+t-1}$, and all possible roots of unity $\{\xi_i\}$ in O_K. Then, we can express β as

$$\beta = \xi\mu \prod_{j=1}^{r} \eta_j^{a_j}$$

for some $\xi \in \{\xi_i\}$, $\mu \in \{\mu_i\}$ and integers $a_1, \ldots, a_r$, where $r = s + t - 1$. One of our primary goals will be to determine all possible ways to express β as a product in the form above. We already know that there are finitely many choices for ξ and μ in this representation. The remaining of this section is devoted to determining an upper bound for the absolute values of the exponents a_i. In turn, we will obtain a finite number of possible representations for the algebraic number β inside K.

Next, take three distinct indices i, j and k, with j, k arbitrary and i defined as an index giving minimal value for the valuations

$$|\beta^{(i)}| = \min_{1 \leq l \leq n} |\beta^{(l)}|.$$

It is not clear beforehand which value of i gives us a minimal valuation of β. One can show that to find the index i, it is enough to look over the indices of the real embeddings

σ_l, that is, $1 \leq i \leq s$ (for a proof refer to Lemma VII.4 in [4]). This remark can be useful in computations, especially when the degree of F is rather large.

The indices i, j and k give rise to a system of three linear equations in X and Y

$$\begin{cases} X - \theta^{(i)}Y = \beta^{(i)} \\ X - \theta^{(j)}Y = \beta^{(j)} \\ X - \theta^{(k)}Y = \beta^{(k)} \end{cases}. \tag{4.2}$$

Solve the first two equations for X and Y, and substitute in the third equation to obtain

$$\beta^{(i)}(\theta^{(j)} - \theta^{(k)}) + \beta^{(j)}(\theta^{(k)} - \theta^{(i)}) + \beta^{(k)}(\theta^{(i)} - \theta^{(j)}) = 0. \tag{4.3}$$

Next, define the following numbers:

$$\alpha_1 = \frac{\xi^{(i)}\mu^{(i)}(\theta^{(j)} - \theta^{(k)})}{\xi^{(j)}\mu^{(j)}(\theta^{(k)} - \theta^{(i)})}, \qquad \alpha_2 = \frac{\xi^{(k)}\mu^{(k)}(\theta^{(i)} - \theta^{(j)})}{\xi^{(j)}\mu^{(j)}(\theta^{(k)} - \theta^{(i)})},$$

and

$$\tau_1 = \prod_{l=1}^{r} \left(\frac{\eta_l^{(i)}}{\eta_l^{(j)}}\right)^{a_l}, \quad \tau_2 = \prod_{l=1}^{r} \left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right)^{a_l}.$$

With the latter definitions, equation (4.3) becomes

$$\alpha_1\tau_1 + \alpha_2\tau_2 + 1 = 0. \tag{4.4}$$

Let us make the following remark: any nonzero complex number $z \in \mathbb{C}$ can be expressed in the form $z = re^{i\theta}$ for some $r > 0$ and $\theta \in \mathbb{R}$. This representation is not unique: $z = re^{i\theta} = re^{i(\theta+2b\pi)}$ for any $b \in \mathbb{Z}$. The *principal argument* of z, denoted $\operatorname{Arg}(z)$, is the unique value of $\theta \in (-\pi, \pi]$ from the representation in the line above. The *principal branch* (also *principal value*) of the complex logarithm is defined as $\operatorname{Log}(z) \stackrel{\text{def}}{=\!=} \log|z| + i\operatorname{Arg}(z)$. The complex logarithm is a multi-valued function, defined as $\log(z) \stackrel{\text{def}}{=\!=} \operatorname{Log}(z) + 2b\pi i$, $b \in \mathbb{Z}$. Each value of b induces a separate *branch* of $\log(z)$. This remark allows us to express

$$\begin{aligned} \Lambda \stackrel{\text{def}}{=\!=} \log(-\alpha_2\tau_2) &= \log(-\alpha_2) + \sum_{l=1}^{r} a_l \log\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right) = \\ &= \operatorname{Log}(-\alpha_2) + 2b_{\alpha_2}\pi i + \sum_{l=1}^{r} a_l \left(\operatorname{Log}\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right) + 2b_l\pi i\right) = \\ &= \operatorname{Log}(-\alpha_2) + \sum_{l=1}^{r} a_l \operatorname{Log}\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right) + 2a_0\pi i, \end{aligned} \tag{4.5}$$

where in the last line we collected all $\mathbb{Z}$-multiples of $2\pi i$ into the single term $2a_0\pi i$, for some $a_0 \in \mathbb{Z}$. We assume that a choice for each b_l and b_{α_2} was made, so that the value of a_0 is also fixed, and hence the expression for Λ is single-valued. To relax notation, from now on we will write the principal branch of the complex logarithm with a lower case letter (namely, "log" will stand for "Log"). Further, all complex logarithms which appear until the end of this thesis will have the principal branch chosen.

Ultimately, using the fact that $\pi i = \log(-1)$, we arrive at the expression

$$\Lambda = \log(-\alpha_2) + \sum_{l=1}^{r} a_l \log\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right) + 2a_0 \log(-1),$$

which is a linear form in complex logarithms of algebraic numbers, also know as a *logarithmic form*. From equation (4.4) we see that if $\alpha_1\tau_1$ is close to 0, then $-\alpha_2\tau_2$ is close to 1, consequently making $\Lambda = \log(-\alpha_2\tau_2)$ close to 0. Lastly, we make the assumption that $\Lambda \neq 0$. We will need this condition in order to use the techniques from lattice reduction, developed in Chapter 3.

Set $A = \max_{i\neq 0} |a_i|$. If we can find a small upper bound on A, we can then loop through all possible a_i and choices for ξ and μ in

$$X - \theta Y + \theta^2 Z_1 + \cdots + \theta^{n-1} Z_{n-2} = \beta = \xi\mu \prod_{j=1}^{r} \eta_j^{a_j}, \quad X, Y, Z_i \in \mathbb{Q}, \quad 1 \leq i \leq n-2$$

and check which cases give rise to coordinates $(X, Y, Z_1, \ldots, Z_{n-2}) = (X, Y, 0, \ldots, 0)$ with $(X, Y) \in \mathbb{Z}^2$. Those integer pairs (X, Y) will correspond to the solutions of (4.1).

Next, rearrange terms in (4.5) to get

$$a_0 2\pi i = \log(-\alpha_2\tau_2) - \log(\alpha_2) - \sum_{l=1}^{r} a_l \log\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right) =$$

$$= \log|-\alpha_2\tau_2| - \log|\alpha_2| - \sum_{l=1}^{r} a_l \log\left|\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right| + i\left(\operatorname{Arg}(-\alpha_2\tau_2) - \operatorname{Arg}(\alpha_2) - \sum_{l=1}^{r} a_l \operatorname{Arg}\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right)\right),$$

which gives

$$|a_0 2\pi| = \left|\operatorname{Arg}(-\alpha_2\tau_2) - \operatorname{Arg}(\alpha_2) - \sum_{l=1}^{r} a_l \operatorname{Arg}\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right)\right| \leq \pi(2 + rA).$$

If $A \geq 2$, then $\frac{A}{2} \geq 1$ makes $|a_0| \leq 1 + r\frac{A}{2} \leq \frac{A}{2} + r\frac{A}{2} = (r+1)\frac{A}{2}$. Now, solving for X in system (4.2) and substituting in the second equation of (4.2) gives us

$$Y(\theta^{(i)} - \theta^{(j)}) = \beta^{(j)} - \beta^{(i)}$$

so then $|Y||\theta^{(i)} - \theta^{(j)}| \leq |\beta^{(j)}| + |\beta^{(i)}| \leq 2|\beta^{(j)}|$, since $|\beta^{(i)}| \leq |\beta^{(j)}|$ from the choice of i. We conclude that, whenever $i \neq j$,

$$|\theta^{(i)} - \theta^{(j)}| \leq |Y||\theta^{(i)} - \theta^{(j)}| \leq 2|\beta^{(j)}|,$$

and then

$$|\beta^{(j)}|^{-1} \leq 2\left|\theta^{(i)} - \theta^{(j)}\right|^{-1} \leq 2\left(\min_{l\neq m}\left|\theta^{(l)} - \theta^{(m)}\right|\right)^{-1} \stackrel{\text{def}}{=\!=} c_1,$$

defining a new constant c_1, not be confused with the one from Theorem 3.2.

Define the constant

$$c_2 \stackrel{\text{def}}{=\!=} \max_{l_1 \neq l_2 \neq l_3 \neq l_1} \left| \frac{\theta^{(l_2)} - \theta^{(l_3)}}{\theta^{(l_3)} - \theta^{(l_1)}} \right|$$

to obtain

$$|\alpha_1 \tau_1| = \left| \frac{\xi^{(i)} \mu^{(i)} (\theta^{(j)} - \theta^{(k)})}{\xi^{(j)} \mu^{(j)} (\theta^{(k)} - \theta^{(i)})} \cdot \frac{\prod_{l=1}^{r} (\eta_l^{(i)})^{a_l}}{\prod_{l=1}^{r} (\eta_l^{(j)})^{a_l}} \right| = \left| \frac{\beta^{(i)} (\theta^{(j)} - \theta^{(k)})}{\beta^{(j)} (\theta^{(k)} - \theta^{(i)])} \right| \leq c_2 \frac{|\beta^{(i)}|}{|\beta^{(j)}|} \leq c_1 c_2 |\beta^{(i)}|.$$

For convenience, set $c_3 \stackrel{\text{def}}{=\!=} c_1 c_2$ and conclude that $|\alpha_1 \tau_1| \leq c_3 |\beta^{(i)}|$. We now turn our attention to the following

Lemma 4.4. *Let K be a number field of degree $n = s+2t$ with $r = s+t-1$ fundamental units $\eta_i \in K$ and let $a_i \in \mathbb{Z}$ for $1 \leq i \leq r$. Define $A = \max_i |a_i|$, set $\epsilon = \prod_{i=1}^{r} \eta_i^{a_i}$ and let $I = \{i_1, \ldots, i_r\} \subset \{1, \ldots, s+t\}$ be an arbitrary subset of size r. Then the following matrix is invertible:*

$$U_I = \begin{pmatrix} \log |\eta_1^{(i_1)}| & \ldots & \log |\eta_r^{(i_1)}| \\ \vdots & & \vdots \\ \log |\eta_1^{(i_r)}| & \ldots & \log |\eta_r^{(i_r)}| \end{pmatrix}.$$

Moreover, there exists an index $t \in \{1, \ldots, s+t\}$, such that

$$\left| \log |\epsilon^{(t)}| \right| \geq A / \|U_I^{-1}\|_\infty,$$

where $\|\|_\infty$ stands for the operator norm induced by the supremum vector norm, i.e., the maximum absolute row sum of a matrix.

Proof. A consequence of the proof of Dirichlet's Unit Theorem, using properties of valuations, is that, up to permuting the indices of the fundamental units η_i, the units satisfy the inequality

$$|\sigma_i(\eta_j)| \begin{cases} < 1, & i \neq j \\ > 1, & i = j, \end{cases} \qquad \text{for} \quad 1 \leq j \leq r \quad \text{and} \quad 1 \leq i \leq r+1$$

(this is Lemma 13.2.12 in [8]). In particular, this tells us that

$$\log |\eta_j^{(i)}| = \log |\sigma_i(\eta_j)| \begin{cases} < 0, & i \neq j \\ > 0, & i = j, \end{cases} \qquad \text{for} \quad 1 \leq j \leq r \quad \text{and} \quad 1 \leq i \leq r+1.$$

Assume for the sake of a contradiction that there is an $\mathbb{R}$-linear dependence relation between the rows of the matrix U_I with weights $x_i \in \mathbb{R}$. Then for any $j \in \{1, \ldots, s+t-1\}$ we have that

$$\sum_{l=1}^{s+t-1} x_{i_l} \log |\sigma_{i_l}(\eta_j)| = 0. \tag{4.6}$$

Multiply this equality by a factor of 2 and define

$$x'_{i_l} = 2x_{i_l} \quad \text{for} \quad 1 \leq i_l \leq s \quad \text{and} \quad x'_{i_l} = x_{i_l} \quad \text{for} \quad s+1 \leq i_l \leq s+t.$$

Furthermore, to simplify notation, introduce

$$a_{i_l,j} = \log|\sigma_{i_l}(\eta_j)| \quad \text{for} \quad 1 \leq i_l \leq s \quad \text{and} \quad a_{i_l,j} = 2\log|\sigma_{i_l}(\eta_j)| \quad \text{for} \quad s+1 \leq i_l \leq s+t.$$

Then $a_{i_l,j} < 0$ for $i_l \neq j$ and > 0 otherwise. With the newly introduced elements in mind, our $\mathbb{R}$-linear dependence (4.6) reads as

$$\sum_{l=1}^{s+t-1} x'_{i_l} a_{i_l,j} = 0. \tag{4.7}$$

Note that $\sum\limits_{i=1}^{s+t} a_{i,j} = \log|N(\eta_j)| = 0$, since the norm $N(\eta_j)$ is ± 1, as η_j is a unit of O_K. We now eliminate the term $a_{s+t,j}$ form (4.7): suppose that x^* is the coefficient in front of $a_{s+t,j}$ in (4.7) ($x^* = 0$ is possible). Then

$$\sum_{l=1}^{s+t-1} x'_{i_l} a_{i_l,j} = 0 = \sum_{i=1}^{s+t} x^* a_{i,j}$$

and shift all nonzero terms to one side, with the $x_{s+t,j}$-term cancelling, to get a linear dependence involving $a_{i,j}$ for $i = 1, \ldots, s+t-1$. Call the weights of this $\mathbb{R}$-linear dependence $z_i = x_i - x^*$:

$$\sum_{i=1}^{s+t-1} z_i a_{i,j} = 0. \tag{4.8}$$

Now, let k correspond to an index making $|z_k|$ the largest among the $|z_i|$'s and assume further that $z_k > 0$ (if not, then rescale all z_i's with a factor -1). By choice of k, it follows that

$$z_i \leq |z_i| \leq |z_k| = z_k, \quad \text{for all possible} \quad i.$$

Then let $j = k$ and rewrite (4.8) in the following way:

$$0 = z_k a_{k,k} + \sum_{i \neq k} z_i a_{i,k} \geq z_k a_{k,k} + z_k \sum_{i \neq k} a_{i,k} = z_k \sum_{i=1}^{s+t-1} a_{i,k},$$

where we use the facts that $z_i \leq z_k$ for all possible i and $a_{i,k} < 0$ for $i \neq k$. Note that $\sum\limits_{i=1}^{s+t-1} a_{i,k}$ is obtained from the sum $\sum\limits_{i=1}^{s+t} a_{i,k} = 0$ by omitting the negative term $a_{s+t,k}$ ($a_{s+t,k} < 0$ as $k \leq s+t-1$). This makes $\sum\limits_{i=1}^{s+t-1} a_{i,k} > 0$, consequently $0 \geq z_k \sum\limits_{i=1}^{s+t-1} a_{i,k} > 0$, which is a contradiction. This shows that U_I is invertible.

Next, from the vector-matrix equation

$$U_I \begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix} = \begin{pmatrix} \log\left(\prod_{i=1}^{r} |\eta_i^{(i_1)}|^{a_i}\right) \\ \vdots \\ \log\left(\prod_{i=1}^{r} |\eta_i^{(i_r)}|^{a_i}\right) \end{pmatrix} = \begin{pmatrix} \log|\epsilon^{(i_1)}| \\ \vdots \\ \log|\epsilon^{(i_r)}| \end{pmatrix}$$

one has

$$\begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix} = U_I^{-1} \begin{pmatrix} \log|\epsilon^{(i_1)}| \\ \vdots \\ \log|\epsilon^{(i_r)}| \end{pmatrix}$$

and taking the supremum norm on both sides gives the inequality

$$A \leq \|U_I^{-1}\|_\infty \cdot \max_{i\in I} |\log|\epsilon^{(i)}|| \leq \|U_I^{-1}\|_\infty \cdot \max_{i=1}^{s+t} |\log|\epsilon^{(i)}||.$$

Ultimately, let $|\log|\epsilon^{(t)}|| = \max_{i=1}^{s+t} |\log|\epsilon^{(i)}||$, to get $|\log|\epsilon^{(t)}|| \geq \dfrac{A}{\|U_I^{-1}\|_\infty}$. □

In view of Lemma 4.4 we obtain an index t and a computable constant c_4 with the property

$$|\log|\epsilon^{(t)}|| \geq c_4 A. \tag{4.9}$$

In [4, Section VII.1] Smart suggests that for c_4 we can take the maximum operator norm $\|U_I^{-1}\|_\infty$ over all possible subsets I (for details on why this is a feasible choice, see proof of Lemma 8.3 in [15]). We next make a choice for a positive constant c_5 which satisfies $c_5 < \dfrac{c_4}{n-1}$. With this new constant, let us investigate the following two possibilities: $|\beta^{(i)}| > e^{-c_5 A}$, or $|\beta^{(i)}| \leq e^{-c_5 A}$, where we a priori made the choice $|\beta^{(i)}| = \min_{1\leq l\leq n} |\beta^{(l)}|$.

From (4.9) we have that either $\log|\epsilon^{(t)}| \geq c_4 A$, or $\log|\epsilon^{(t)}| \leq -c_4 A$, that is, either $|\epsilon^{(t)}| \geq e^{c_4 A}$, or $|\epsilon^{(t)}| \leq e^{-c_4 A}$. With this in mind, split the case $|\beta^{(i)}| > e^{-c_5 A}$ into two sub cases:

Case A: $|\beta^{(i)}| > e^{-c_5 A}$ **and** $|\epsilon^{(t)}| \geq e^{c_4 A}$

Recall that $\beta^{(i)} = \xi^{(i)}\mu^{(i)}\epsilon^{(i)}$ and note that $\prod_{l=1}^{n} |\beta^{(l)}| = |\prod_{l=1}^{n} \sigma_l(\beta)| = |N(\beta)| = |m|$. Therefore,

$$|\beta^{(t)}| = \frac{|m|}{\prod_{l\neq t} |\beta^{(l)}|} \leq \frac{|m|}{|\beta^{(i)}|^{n-1}} = |m| \cdot |\beta^{(i)}|^{-(n-1)} < |m| e^{(n-1)c_5 A}.$$

Then $e^{c_4 A} \leq |\epsilon^{(t)}| = \dfrac{|\beta^{(t)}|}{|\mu^{(t)}|} \leq c_6 |\beta^{(t)}|$, where $c_6 \overset{\text{def}}{=\!=} \max_{1\leq l\leq n} \dfrac{1}{|\mu^{(l)}|}$. Further, $e^{c_4 A} \leq c_6|\beta^{(t)}| < c_6|m|e^{(n-1)c_5 A}$. Now, after taking the logarithm on both sides, one has

$$c_4 A < (n-1)c_5 A + \log(c_6|m|),$$

from which

$$A < \frac{\log(c_6|m|)}{c_4 - (n-1)c_5} \stackrel{\text{def}}{=\!=} A_1.$$

Case B: $|\beta^{(i)}| > e^{-c_5 A}$ **and** $|\epsilon^{(t)}| \leq e^{-c_4 A}$
Let $c_7 \stackrel{\text{def}}{=\!=} \min\limits_{1\leq l\leq n} \dfrac{1}{|\mu^{(l)}|}$. In this case

$$e^{-c_4 A} \geq |\epsilon^{(t)}| = \frac{|\beta^{(t)}|}{|\mu^{(t)}|} \geq \min_{1\leq l\leq n} \frac{1}{|\mu^{(l)}|} \cdot |\beta^{(t)}| = c_7|\beta^{(t)}| > c_7 e^{-c_5 A}.$$

Take the logarithm on both sides of the above inequality to get the following bound on A: $-c_4 A > \log(c_7) - c_5 A$, so $A(c_5 - c_4) > \log(c_7)$. But c_5 was chosen to make $c_4 > (n-1)c_5$, so $c_5 - c_4 < 0$, hence

$$A < \frac{\log(c_7)}{c_5 - c_4} \stackrel{\text{def}}{=\!=} A_2.$$

Case C: $|\beta^{(i)}| \leq e^{-c_5 A}$
We have seen before that $|\alpha_1\tau_1| \leq c_3|\beta^{(i)}|$, so $|\alpha_1\tau_1| \leq c_3 e^{-c_5 A}$. We now make use of the following

Lemma 4.5. *Suppose that* $0 < a < 1$ *and let* $\Delta \in \mathbb{C}\backslash\{0\}$ *satisfy the inequality* $|\Delta - 1| \leq a$. *Then* $|\log \Delta| \leq \dfrac{-\log(1-a)}{a} \cdot |\Delta - 1|$.

Proof. Recall the Newton-Mercator series for the principal branch of the complex logarithm

$$\log(1+u) = \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i} u^i,$$

which converges whenever $|u| < 1$. So suppose that $|\Delta - 1| < a < 1$ and set $u = \Delta - 1$ to get

$$\begin{aligned} |\log \Delta| &= \left| \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i} (\Delta - 1)^i \right| = |\Delta - 1| \left| \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i} (\Delta - 1)^{i-1} \right| = \\ &= |\Delta - 1| \left| \sum_{i=1}^{\infty} \frac{(-1)^{i-1}}{i} (\Delta - 1)^{i-1} \right| \leq |\Delta - 1| \sum_{i=1}^{\infty} \frac{a^{i-1}}{i} = \frac{|\Delta - 1|}{a} \sum_{i=1}^{\infty} \frac{a^i}{i} = \\ &= -\frac{\log(a-1)}{a} \cdot |\Delta - 1|, \end{aligned}$$

where we used the Taylor-expansion $-\log(1-a) = \sum\limits_{i=1}^{\infty} \dfrac{a^i}{i}$. □

Now, for any $z \in \mathbb{C}\backslash\{0\}$ with $|z-1| < \frac{1}{2}$, apply Lemma 4.5 with $a = 1/2$ to get

$$|\log(z)| \leq \frac{-\log(1-1/2)}{1/2}|z-1| = 2|z-1|\log(2) < 2|z-1|. \tag{4.10}$$

In order to make use of the above lemma, we will need $c_3 e^{-c_5 A} \leq \frac{1}{2}$, which is equivalent to $A \geq \dfrac{\log(2c_3)}{c_5} \overset{\text{def}}{=\!=} A_3$. So, if $A \geq A_3$, then $|e^{\Lambda}-1| = |e^{\log(1+\alpha_1\tau_1)}-1| = |\alpha_1\tau_1| \leq c_3 e^{-c_5 A} \leq \frac{1}{2}$, and according to (4.10), $|\log e^{\Lambda}| = |\Lambda| \leq 2|e^{\Lambda}-1| = |\alpha_1\tau_1| \leq 2c_3 e^{-c_5 A}$. Ultimately, we find that

$$|\Lambda| \leq 2c_3 e^{-c_5 A}. \tag{4.11}$$

At this stage we recall that

$$\Lambda = \log(-\alpha_2) + \sum_{l=1}^{r} a_l \log\left(\frac{\eta_l^{(k)}}{\eta_l^{(j)}}\right) + 2a_0\log(-1)$$

is a logarithmic form and we are in position to apply results from Baker's theory. The main result to help us in this situation is the next theorem, in the formulation of which we use the convention on minimal polynomials to have integral coefficients with a positive leading coefficient.

Theorem 4.6 (Baker-Wüstholz)**.** *Let $\alpha_1, \ldots, \alpha_n$ denote algebraic numbers not equal to 0 or 1 ($n \geq 2$). Let $K = \mathbb{Q}(\alpha_1, \ldots, \alpha_n)$ and set $d = \deg[K : \mathbb{Q}]$. Define the following "heights": the* height *$H(\alpha)$ of $\alpha \in K$ is defined by $H(\alpha)^{d_\alpha} = a_0 \prod_i \max(1, |\alpha^{(i)}|)$, where a_0 is the leading coefficient of the minimal polynomial of α, d_α-its degree, and $\alpha^{(i)}$-the conjugates of α; the* logarithmic height *is taken to be $h(\alpha) = \log H(\alpha)$; the* modified logarithmic height *is taken to be*

$$h_m(\alpha) = \max\left\{h(\alpha), \frac{|\log(\alpha)|}{d}, \frac{1}{d}\right\}.$$

Let $b_1, \ldots, b_n$ be integers such that

$$\Lambda = b_1\log(\alpha_1) + \cdots + b_{n-1}\log(\alpha_{n_1}) + b_n\alpha_n \quad \textit{is nonzero.}$$

Then, if $B = \max\{|b_1|, \ldots, |b_n|\} \geq 3$, we have the inequality

$$\log|\Lambda| > -c_1 h_m(\alpha_1)\ldots h_m(\alpha_n)\log(B) \quad \textit{with}$$

$$c_1 = 18(n+1)!n^{n+1}(32d)^{n+2}\log(2nd).$$

Proof. Refer to the paper by A. Baker and G. Wüstholz [17]. □

We have shown that if $A \geq 2$ one has $|a_0| \leq \dfrac{(r+1)A}{2}$. Then, let $A \geq 3$ to get that

$$\frac{r+1}{2} \geq 1 \implies \frac{r+1}{2}A \geq \begin{cases} A = \max\limits_{i\neq 0}|a_i| \\ |a_0| \end{cases} \implies \frac{r+1}{2}A \geq \max_{i=0}^{r}|a_i| \overset{\text{def}}{=\!=} B.$$

Apply Theorem 4.6 (which is made possible under the condition $A \geq 3$) to get a positive constant c_8 such that

$$\log|\Lambda| > -c_8 \log(B) > -c_8 \log\left(\frac{r+1}{2}A\right). \tag{4.12}$$

Application of the Theorem is justified under the assumption that $\Lambda \neq 0$, which we have agreed to be the case for our linear forms.

From inequality (4.11), after taking the logarithm on both sides, and combining with (4.12), one has

$$\log|\Lambda| \leq \log(2c_3) - c_5 A \iff c_5 A \leq \log|\Lambda| + \log(2c_3) < c_8 \log(\frac{r+1}{2}A) + \log(2c_3),$$

hence

$$A \leq \frac{1}{c_5}\left(c_8 \log(A) + \log(2c_3) + c_8 \log\left(\frac{r+1}{2}\right)\right).$$

We shall next make use of a result due to Pethö and de Weger.

Lemma 4.7 (Pethö and de Weger)**.** *Let $a, b \geq 0$, $h \geq 1$ and $x \in \mathbb{R}$ be the largest solution to the equation $x = a + b(\log x)^h$. If $b > (e^2/h)^h$, then*

$$x < 2^h \left(a^{1/h} + b^{1/h} \log(h^h b)\right)^h, \quad \textit{and if} \quad b \leq (e^2/h)^h, \quad \textit{then} \quad x \leq 2^h \left(a^{1/h} + 2e^2\right)^h.$$

Proof. Refer to Lemma B.1 in [4]. □

In our situation, we apply Lemma 4.7 with $b = \dfrac{c_8}{c_5}$, $a = \dfrac{1}{c_5}\log(2c_3) + \dfrac{c_8}{c_5}\log\left(\dfrac{r+1}{2}\right)$ and $h = 1$ to get

$$A \leq \begin{cases} 2\left(\frac{1}{c_5}\log(2c_3) + \frac{c_8}{c_5}\log\left(\frac{r+1}{2}\right) + \frac{c_8}{c_5}\log(\frac{c_8}{c_5})\right), \text{if } b > e^2 \\ 2\left(\frac{1}{c_5}\log(2c_3) + \frac{c_8}{c_5}\log\left(\frac{r+1}{2}\right) + 2e^2\right), \text{if } b \leq e^2. \end{cases}$$

In our concrete calculations later on, we find ourselves in the case of $b > e^2$, due to the fact that c_8 will be rather large (in fact, as large as $c_8 \approx 10^{22}$). With that we have just obtained an upper bound A_4 on A:

$$A \leq A_4 \overset{\text{def}}{=\!=} \frac{2}{c_5}\left(\log\left(2c_3 + c_8 \log\left(\frac{r+1}{2}\right) + c_8 \log\left(\frac{c_8}{c_5}\right)\right)\right).$$

In all of our examples, the upper bound A_4 will be astronomically large and inconvenient for carrying out enumeration at this stage. Fortunately enough, (4.11) and Lemma 3.4 allow us to deduce a significantly smaller bound on A and let us call this newly determined bound A_5.

Lastly, capture the largest of the upper bounds obtained from **Cases A, B, C** and account for the requirements $A \geq 3$ and $A \geq A_3$, in defining $A_6 = \max\{3, A_1, A_2, A_3, A_5\}$.

To solve our equation $F(X,Y) = m$ it now remains loop over all $(2A_6+1)^r$ possible values for the integers a_i in the expression

$$X - \theta Y = \mu\xi \prod_{i=1}^{r} \eta_i^{a_i}$$

for each of the possible choices for μ and ξ. Those cases which give rise to $(X,Y) \in \mathbb{Z}^2$ will correspond to the solutions of the Thue equation (4.1). We shall illustrate how the method works in practice on a concrete example in the section to follow.

4.2 Applying the Method

4.2.1 The Equation $4X^4 - 3Y^4 = \pm 61$

In this section we demonstrate how the method of Section 4.1 can be applied to solve the Diophantine equation $4X^4 - 3Y^4 = \pm 61$. In the manner suggested at the beginning of Section 4.1, we multiply both sides of this equation with 4^3 and reduce the original Diophantine problem to solving the Diophantine equation

$$F(X,Y) = X^4 - 192Y^4 = \pm 3904.$$

Let $F(X,Y) = X^4 - 192Y^4$ and let θ be the real root $\theta \approx -3.72242$ of the polynomial $F(X,1) = X^4 - 192$. In $K = \mathbb{Q}(\theta)$ we find the fundamental units

$$\eta_1 = -\frac{1}{2} - \frac{1}{4}\theta - \frac{1}{16}\theta^2 \quad \text{and} \quad \eta_2 = 2 + \frac{1}{8}\theta^2.$$

The only roots of unity in K are ± 1 - a consequence of the fact that K has a real embedding into $\mathbb{C}$. Next we find a complete set of representatives of the elements of O_K of norm ± 3904, up to associates, and such elements are

$$\mu_1 = -8 - \theta, \quad \mu_2 = -4 - \frac{1}{2}\theta^2 - \frac{1}{8}\theta^3, \quad \mu_3 = -2 + \frac{1}{2}\theta - \frac{1}{2}\theta^2 - \frac{1}{16}\theta^3,$$

and $\mu_4 = 4 + \dfrac{3}{2}\theta + \dfrac{1}{4}\theta^2 - \dfrac{1}{16}\theta^3$. Start with $\mu = \mu_1$. We are interested in elements

$$X - \theta Y = \pm\mu \cdot \eta_1^{a_1}\eta_2^{a_2}.$$

Computation of the constants c_i, $1 \le i \le 4$, gives

$$c_1 = 0.379918, \quad c_2 = 1.41421, \quad c_3 = 0.537285, \quad c_4 = 2.40546.$$

We make the choice $c_5 = c_4/4 = 0.601364$. The vales of c_6 and c_7 are both equal to 0.233777 and in view of **Cases A** and **B** we get the upper bounds $A_1 = 4.4191$ and $A_2 = 0.805606$, respectively.

We move onto **Case C**. The following procedure has to be performed for the values $i = 1$ and $i = 2$ separately.

First, let $i = 1$. We make a choice $j = 2$ and $k = 3$, and numerically approximate

$$\alpha_2 \overset{\text{def}}{=\!=} \pm\frac{\theta^{(1)} - \theta^{(2)}}{\theta^{(3)} - \theta^{(1)}} \approx \pm\,(-1 - i), \quad s_1 \overset{\text{def}}{=\!=} \frac{\eta_1^{(3)}}{\eta_1^{(2)}} \approx -0.159375 - 0.405204i,$$

$$s_2 \overset{\text{def}}{=\!=} \frac{\eta_2^{(3)}}{\eta_2^{(2)}} \approx 0.0717968 + 0.i.$$

Now consider the linear form

$$\Lambda = \log(-\alpha_2) + a_1 \log(s_1) + a_2 \log(s_2) + 2a_0 \log(-1).$$

We need to consider both values of α_2 which, differ by a sign. First, let $\alpha_2 = \dfrac{\theta^{(1)} - \theta^{(2)}}{\theta^{(3)} - \theta^{(1)}}$. The modified heights, as defined in Theorem 4.6, are

$$h_m(\alpha_2) = 0.346574, \quad h_m(s_1) = 0.415721, \quad h_m(s_2) = 0.831443, \quad h_m(-1) = 0.392699,$$

which allow us to compute the constant $c_8 = 1.21804 \cdot 10^{20}$. Then, if $A \geq A_3 = \dfrac{\log(2c_3)}{c_5} = 0.119596$, it follows that

$$A \leq A_4 = \frac{2}{c_5}\left(\log\left(2c_3 + c_8 \log\left(\frac{r+1}{2}\right) + c_8 \log\left(\frac{c_8}{c_5}\right)\right)\right) = 1.91053 \cdot 10^{22}.$$

With the latter upper bound and the inequality $|\Lambda| \leq 2c_3 e^{-c_5 A}$, we will deduce a smaller upper bound on A, using the technique from Section 3.2. To achieve this, make the choice $C = 10^{50}$ and construct the approximation lattice

$$B = \begin{pmatrix} 1 & 0 & 0 \\ [C\Re(\log(s_1))] & [C\Re(\log(s_2))] & 0 \\ [C\Im(\log(s_1))] & [C\Im(\log(s_2))] & [C2\pi] \end{pmatrix},$$

the vector

$$\vec{y} = \begin{pmatrix} 0 \\ -[C\Re(-\alpha_2)] \\ -[C\Im(-\alpha_2] \end{pmatrix},$$

and apply Lemma 3.4 to deduce that $A \leq 97.1803$. Next, set a smaller value $C = 10^{10}$, use the bound $A \leq 97$ and repeat the same procedure to deduce that $A \leq 19.4364$. Further, reducing to $C = 10^5$ we get $A \leq 10.3902$, which is small enough for direct enumeration. The same bound on A is obtained when setting $\alpha_2 = -\dfrac{\theta^{(1)} - \theta^{(2)}}{\theta^{(3)} - \theta^{(1)}}$.

By complete analogy we treat the case $i = 2$ with the choice of $j = 1$ and $k = 3$. In this instance, for $C = 10^5$ and both sign-alternative choices of α_2, we obtain the same reduced bound $A \leq 10.3902$.

Ultimately, in the case of $\mu = \mu_1$, we check all possible $(2 \cdot 10 + 1)^2$ numbers of the form

$$X - \theta Y = \pm\mu \cdot \eta_1^{a_1} \eta_2^{a_2},$$

for values of $a_{1,2} \in [-10, 10] \cap \mathbb{Z}$, and capture those, which give rise to integer pairs (X, Y). The obtained integer solutions to $X^4 - 192Y^4 = \pm 3904$ are $\pm(X, Y) = (-8, 1)$.

Now we need to perform the exact same process for the remaining choices of μ. We spare details and directly give the corresponding solution pairs: after setting $\mu = \mu_2$ and $\mu = \mu_3$ no integer solutions arise; setting $\mu = \mu_4$ gives us solutions $\pm(X, Y) = (-8, -1)$.

In turn, after rescaling the X-component with a factor $1/4$ from all of the obtained integer solutions, we get all integer solutions to the original equation $4X^4 - 3Y^4 = \pm 61$, namely $\pm(X, Y) = (-2, -1), (-2, 1)$.

4.2.2 Choice of Units

In this subsection we give a short analysis on how the choice of fundamental units can affect the reduced bounds we obtain from the method of Tzanakis and de Weger. For illustrative purposes, we restrict the discussion to the Thue equation $X^4 - 2Y^4 = \pm 49$.

First, we note that fundamental units, generating the free part of the unit group O_K^*, are unique up to coordinate transformation and multiplication by roots of unity, i.e. a system of fundamental units is, up to torsion, unique up to $GL_n(\mathbb{Z})$ action. A special case of this transformation is replacing a unit η_i from a system of fundamental units $\{\eta_1, \ldots, \eta_r\}$ with its inverse η_i^{-1}. The resulting set will still comprise of a system of fundamental units. We will now investigate how the choice of units affects the bounds obtained in the method of Section 4.1. Usually, the largest upper bounds on A occur in **Case C**, and can be summarised as:

- $A \geq A_3 \implies A \leq A_4 = \dfrac{8}{c_4}\left(\log\left(2c_3 + c_8 \log\left(\frac{r+1}{2}\right) + c_8 \log\left(\frac{c_8}{c_5}\right)\right)\right)$, which we shall call the "direct bound";
- $A \leq A_5 = \dfrac{4}{c_4}\left(\log(2C \cdot c_3) - G\right)$, for a computable constant G given from the step in which we apply Lemma 3.4, which we shall call the "reduced bound".

For the particular Thue equation $X^4 - 2Y^4 = \pm 49$, we set $\theta \approx -1.18921$ - one of the roots of $X^4 - 2$, and in $K = \mathbb{Q}(\theta)$ we obtain four representatives of norm ± 49, from which we pick the one $\mu = -1 - \theta + \theta^2 - \theta^3$. Further, we set $i = 1$, $j = 3$ and $k = 2$. To investigate possible dependencies on the choice of fundamental units, we provide a table comparing fundamental units, values of the constant c_4 and direct and reduced bounds, after a choice for $C = 10^{50}$ was made.

Fundamental Units	c_4	Direct Bound A_4	Reduced Bound A_5
$\eta_1 = 1-\theta \approx 2.18921$ $\eta_2 = -1-\theta \approx 0.189207$	1.95144	$2.56854 \cdot 10^{22}$	120.682
$\eta_1 = (1+\theta)^{-1} \approx -5.28521$ $\eta_2 = 1-\theta \approx 2.18921$	1.95144	$2.56854 \cdot 10^{22}$	119.969
$\eta_1 = (1+\theta)^{-1} \approx -5.28521$ $\eta_2 = (-1+\theta)^{-1} \approx -0.456786$	1.95144	$2.56854 \cdot 10^{22}$	120.007
$\eta_1 = -1-\theta \approx 0.189207$ $\eta_2 = (-1+\theta)^{-1} \approx -0.456786$	1.95144	$2.56854 \cdot 10^{22}$	119.969

We can see that, with the astronomical choice $C = 10^{50}$, different choices of units do not affect the value of c_4 and produce the same direct bound on A. The reduced bound, on the other hand, varies, yet differences appear insignificant.

To further study the influence of the choice of units, we perform successive iterations of the method, gradually reducing the value of C, in the manner performed in Section 4.2.1. With the choice of units $\eta_1 = (-1-\theta)^{-1}$, $\eta_2 = 1-\theta$, we are able to reduce the bound to $A \leq 80$, with least possible value of $C = 10^{19}$. Note that one can proceed with taking for C multiples of the form $n \cdot 10^k$ for $k \leq 18, n \in \{1, \ldots, 9\}$, yet we shall restrict ourselves only to powers of 10 for illustration. On the other hand, with the choice of units $\eta_1 = -1-\theta$, $\eta_2 = 1-\theta$, we are able to reduce the bound on A to $A \leq 11$, with ultimate value of $C = 10^4$.

The explanation of our "success" in the second reduction is the following: let B_{LLL} denote the matrix of the LLL-reduced basis for the approximation lattice associated to the form Λ, and $\vec{y}$ - the vector capturing the inhomogeneity term of the linear form (analogously to Section 4.2.1). Call $\vec{b_1}$ the first column vector of B_{LLL} and A' - an upper bound on A (either the direct bound or one obtained from previous reduction step). In order to execute the (next) reduction step, we need to satisfy the inequality "$c_4^2 \geq T^2 + S$" cited from Lemma 3.4. In our current context this inequality reads as:

$$\|\vec{b_1}\|^2 \cdot \{\sigma(\vec{3})\} \geq (A')^2 + (1+3A')^2 \cdot 2^{-1}, \tag{4.13}$$

where $\vec{\sigma} = (B_{LLL})^{-1} \vec{y}$, and in particular, we need the first vector of the LLL-reduced basis to have "large" Euclidean norm, relative to the value of A'. To showcase this, for the first choice of units $\eta_1 = (-1-\theta)^{-1}$, $\eta_2 = 1-\theta$ with reduced bound $A' = 80$, we set $C = 10^{18}$ and apply the method. In this situation we have $\|\vec{b_1}\| = 173.012$ and fail to satisfy inequality (4.13). In contrast, with the same bound A' and choice $C = 10^{18}$, the units $\eta_1 = -1-\theta$, $\eta_2 = 1-\theta$ give much larger $\|\vec{b_1}\| = 1.89857 \cdot 10^9$, satisfying (4.13), and allowing us to drop the bound to $A \leq 44.4498$.

In view of the above analysis, we can draw the following conclusion: the choice of units can have an effect on the efficiency of the method of Tzanakis and de Weger for Thue equations. In particular, different units give rise to different norms of the smallest vector inside the respective approximation lattice. In turn some choices of units will do a better job at reducing the upper bound on A than others. To maximise efficiency of the method, we suggest the following strategy:

1. Set a value for C and a bound on A - either the direct bound or the one obtained from the first reduction step.
2. For different choices of units, compute the norm $\|\vec{b_1}\|$ and choose those fundamental units maximizing this norm.
3. With the maximizing units from step 2., successively apply the method to deduce smaller upper bounds on A, until no further reduction is possible.

In step 2. above, computations of $\|\vec{b_1}\|$ take fractions of a second. On the other hand, the final enumeration step over the possible values of a_i will be more computationally demanding. To illustrate, when performing enumeration for $X^4 - 2Y^4 = \pm 49$, with the choice of μ made above: units $\eta_1 = (-1-\theta)^{-1}$, $\eta_2 = 1-\theta$, with ultimate bound $A \leq 80$, take more than two hours for enumeration, after which we aborted the process; the more careful choice $\eta_1 = -1-\theta$, $\eta_2 = 1-\theta$, with ultimate bound $A \leq 11$, completes enumeration in 4.30 seconds. For computational details see `TetstingUnitsPart1.nb` and `TetstingUnitsPart2.nb` in the GitHub link below.

We can not say a priori which units will produce the largest possible $\|\vec{b_1}\|$. Further study on how the units relate to the norm $\|\vec{b_1}\|$ could be possible through in-depth analysis of the underlying LLL-reduction algorithm of the respective algebra computer package. We leave this as a suggestion for possible future work.

4.2.3 `Mathematica` Implementation

After implementing the method of Tzanakis and de Weger in `Mathematica`, we solved the following fourth degree Thue equations:

$F(X,Y) = \pm m$	Solutions (X,Y)
$X^4 - 2Y^4 = \pm 49$	$\pm(3,-2), \pm(3,2)$
$X^4 - 2Y^4 = \pm 46$	$\pm(6,5), \pm(6,-5)$
$X^4 + 3X^3Y + X^2Y^2 + 2Y^4 = \pm 7$	$\pm(1,1), \pm(5,-2)$
$X^4 - X^3Y + 3X^2Y^2 - 2Y^4 = \pm 17$	$\pm(-9,13), \pm(5,6), \pm(1,-2)$
$X^4 - X^3Y + X^2Y^2 - 3XY^3 - 7Y^4 = \pm 1$	$\pm(1,0), \pm(2,1), \pm(1,-1)$
$3X^4 - 2Y^4 = \pm 1$	$\pm(1,-1), \pm(1,1)$
$4X^4 - 3Y^4 = \pm 61$	$\pm(2,1), \pm(-2,1)$

Our implementation for each of the above Diophantine problems can be found in the following link: https://github.com/Zary4545/Thesis-Mathematica.git. We provide two additional notebooks in which we carried out analysis for the choice of units.

Remarks on implementation

In `Mathematica` there are built-in methods for computing *fundamental units* of a number field, *representatives of norm* $\pm m \in \mathbb{Z}\backslash\{0\}$ and *roots of unity*, respectively in the commands `NumberFieldFundamentalUnits`, `NumberFieldRootsOfUnity` and `NumberFieldNormRepresentatives`. We can also find minimal polynomials of algebraic numbers (`MinimalPolynomial`) and a primitive element of a finite degree extension of $\mathbb{Q}$ (`ToNumberField[..., All]`). These methods aid us in computing the modified heights of Theorem 4.6. One of the methods we had to define ourselves is `Embed[a_,i_]`, which takes input element $a \in K$ and index i, upon which it gives us the image $\sigma_i(a) = a^{(i)} \in K^{(i)}$. The method `EmbedQ` is just identity map and was defined only to aid us in adapting the code from one Thue equation to another.

The method `LatticeReduce` computes an LLL-reduced lattice basis. To carry out the computations of Lemma 3.4 we defined the method `GramSchmidt`, which computes an orthogonal basis via the Gram-Schmidt process without normalizing the basis vectors.

Remarks on computation

Our implementation of the method is not automated and requires manual intervention. We split the method of Tzanakis and de Weger into separate blocks, each one of which needs to be executed separately.

One matter of concern in our implementation occurs in the final enumeration step, when we compute all numbers of the particular form $X - \theta Y \in K$ and look for integer pairs (X, Y). For some equations we were unable do drop the bound on A to something "too small". For the equation $X^4 - X^3Y + X^2Y^2 - 3XY^3 - 7Y^4 = \pm 1$ we set $\theta = -1.07805$ and find a single representative $\mu = 1$ of norm ± 1 and two fundamental units $\eta_1 = -1 - \theta$, $\eta_2 = 2 - \theta$. After successive applications of the reduction technique, our ultimate bound was $A \leq 30$. This forced us to loop over "large" values for the exponents of the units during the final enumeration. In turn, when type `AlgebraicNumber` was raised to high powers, `Mathematica` gave us warning messages for encountering infinite expressions. This raised the concern that possible solutions could remain undetected during the enumeration step. This brings us to our next note.

Comparison with `Reduce`

An implemented `Mathematica` solver for Diophantine equations is found in the function `Reduce["expr", "vars", Integers]`. In order to check if any solutions are missed in our implementations, we solved each Thue equation from the table using `Reduce`. We found the same solutions as those obtained from our own implementation. However, it could still be that all solutions appeared for small values of the exponents of the units, yet no further reduction on the bound of A was possible.

Lastly, we make a note about efficiency. The solver `Reduce[expr, vars, dom]` is fully automated, unlike our implementation. `Reduce` is also noticeably faster. The final enumeration of the example $X^4 - X^3Y + X^2Y^2 - 3XY^3 - 7Y^4 = \pm 1$, mentioned above, took 70.8265 minutes to complete, while `Reduce` completely solves the problem in 2.83 seconds.

5 The Method of Bilu and Hanrot

In this chapter we give a short derivation of a method for solving Thue equation, due to Yuri Bilu and Guillaume Hanrot. Section 5.1 is dedicated to the derivation, while Section 5.2 discusses the final enumeration procedure. Current exposition follows the outline of Section VII.3 in [4]. Main references include the papers by Bilu and Hanrot [10] and [14], as well as the paper by Tzanakis [11].

5.1 Brief Exposition

One modification of the method of Tzanakis and de Weger is due to Yuri Bilu and Guillaume Hanrot. In their approach we again reduce the Diophantine problem to determining an upper bound on the exponents of the units. Their method will turn out to have certain computational advantages compared to the method of Chapter 4. We now start the exposition.

Follow the steps of Section 4.1 until Lemma 4.4, introducing the matrix U_I. Choose the index i such that

$$|\beta^{(i)}| = \min_{1\le l\le r+1} |\beta^{(l)}|$$

and let the subset I be formed by $\{i\} = \{1,\dots,s+t\}\backslash I$. Denote $(u_{i,j}) = U_I^{-1}$. Recall that in $X-\theta Y = \xi\mu\epsilon$, $\xi\in\mathbb{C}$ was a root of unity, so $|\xi|=1$ and thus $|X-\theta Y| = |\mu\epsilon|$. With that remark, from the vector-matrix equation

$$U_I \begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix} = \begin{pmatrix} \log|\epsilon^{(i_1)}| \\ \vdots \\ \log|\epsilon^{(i_r)}| \end{pmatrix} = \begin{pmatrix} \log\left|\frac{X-\theta^{(i_1)}Y}{\mu^{(i_1)}}\right| \\ \vdots \\ \log\left|\frac{X-\theta^{(i_r)}Y}{\mu^{(i_r)}}\right| \end{pmatrix}, \quad \text{we get} \quad \begin{pmatrix} a_1 \\ \vdots \\ a_r \end{pmatrix} = U_I^{-1} \begin{pmatrix} \log\left|\frac{X-\theta^{(i_1)}Y}{\mu^{(i_1)}}\right| \\ \vdots \\ \log\left|\frac{X-\theta^{(i_r)}Y}{\mu^{(i_r)}}\right| \end{pmatrix},$$

and so for all $k=1,\dots,r$ we have that

$$a_k = \sum_{j=1}^{r} u_{k,j} \log\left|\frac{X-\theta^{(i_j)}Y}{\mu^{(i_j)}}\right| = \sum_{j=1}^{r} u_{k,j}\log\left|Y\cdot\frac{X/Y-\theta^{(i_j)}}{\mu^{(i_j)}}\right| = \tag{5.1}$$

$$= \left(\sum_{j=1}^{r} u_{k,j}\right)\log|Y| + \sum_{j=1}^{r} u_{k,j}\log\left|\frac{\theta^{(i)}-\theta^{(i_j)}}{\mu^{(i_j)}}\right| + \sum_{j=1}^{r} u_{k,j}\log\left|\frac{X/Y-\theta^{(i_j)}}{\theta^{(i)}-\theta^{(i_j)}}\right| = \tag{5.2}$$

$$= \delta_k\log|Y| + \lambda_k + \sum_{j=1}^{r} u_{k,j}\log\left|\frac{X/Y-\theta^{(i_j)}}{\theta^{(i)}-\theta^{(i_j)}}\right|, \tag{5.3}$$

after setting

$$\delta_k \overset{\text{def}}{=\!=} \sum_{j=1}^{r} u_{k,j}, \quad \lambda_k \overset{\text{def}}{=\!=} \sum_{j=1}^{r} u_{k,j} \log \left| \frac{\theta^{(i)} - \theta^{(i_j)}}{\mu^{(i_j)}} \right|, \quad \forall k \in \{1, \ldots, r\}.$$

Next, we introduce the following result:

Lemma 5.1. *Define $g(X) = F(X, 1)$. Let t be the number of pairs of complex-conjugate nonreal embeddings from K into $\mathbb{C}$. Define*

$$Y_0 = \begin{cases} (2^{n-1}|m|) \left(\min\limits_{1 \le i \le t} |g'(\theta^{(s+i)})| \cdot \min\limits_{1 \le i \le t} |\Im(\theta^{(s+i)})| \right)^{-1}, & \text{if } t \ge 1 \\ 1, \quad \text{if } t = 0 \end{cases},$$

$c_1' = \dfrac{2^{n-1}|m|}{\min\limits_{1 \le i \le s} |g'(\theta^{(i)})|}$, $\quad c_2' = \min\limits_{1 \le i < j \le n} |\theta^{(i)} - \theta^{(j)}|$, *and let* $Y_1 = \max\left\{ Y_0, \left((n+1) \frac{c_1'}{c_2'} \right)^{1/n} \right\}$.
Let (X, Y) be an integer solution to $F(X, Y) = m$ and call i_0 the index for which $|\beta^{(i_0)}| = \min\limits_{1 \le l \le r+1} |\beta^{(l)}|$. Then:

1. $|Y| > Y_0 \implies \left| \frac{X}{Y} - \theta^{(i_0)} \right| \le \frac{c_1'}{|Y|^n}$,
2. $|Y| > Y_1 \implies \frac{n+2}{n+1} |\theta^{(i_0)} - \theta^{(i)}| \ge \left| \frac{X}{Y} - \theta^{(i)} \right| \ge \frac{n}{n+1} |\theta^{(i_0)} - \theta^{(i)}|, \ (i \ne i_0)$.

Proof. See the proof of Proposition 2.2.1 in [10] and/or Lemma 1.1 in [11]. □

So now if $|Y| > Y_1$, then

$$\log \left| \frac{X/Y - \theta^{(i_j)}}{\theta^{(i)} - \theta^{(i_j)}} \right| \le \log \left(\frac{n+2}{n+1} \right)$$

and we obtain the inequality

$$\left| \sum_{j=1}^{r} u_{k,j} \log \left| \frac{X/Y - \theta^{(i_j)}}{\theta^{(i)} - \theta^{(i_j)}} \right| \right| \le \sum_{j=1}^{r} |u_{k,j}| \log \left| \frac{n+2}{n+1} \right|.$$

A not too difficult exercise is showing that $\log \left(\dfrac{n+2}{n+1} \right) \le \dfrac{1}{n}, \forall n \in \mathbb{N}$ and this allows us to rewrite (5.3) as

$$|a_k| \le |\delta_k| \log |Y| + |\lambda_k| + \frac{1}{n} \sum_{j=1}^{r} |u_{k,j}|, \quad \forall k \in \{1, \ldots, r\}. \tag{5.4}$$

Define $c_{10} \overset{\text{def}}{=\!=} \max\limits_{1 \le k \le r} |\delta_k|$ and $c_{11} \overset{\text{def}}{=\!=} \max\limits_{1 \le k \le r} \left(|\lambda_k| + \frac{1}{n} \sum\limits_{j=1}^{r} u_{k,j} \right)$ so that $A \overset{\text{def}}{=\!=} \max\limits_{i=1}^{r} |a_i| \le c_{10} \log |Y| + c_{11}$. From the last inequality we obtain

$$|Y|^{-1} \le \exp\left(\frac{c_{11}}{c_{10}} \right) \cdot \exp\left(-\frac{1}{c_{10}} A \right),$$

and setting $c_{12} \overset{\text{def}}{=\!=} e^{c_{11}/c_{10}}$, $c_{13} \overset{\text{def}}{=\!=} 1/c_{10}$, we obtain the inequality $|Y|^{-1} \le c_{12} e^{-c_{13} A}$.

Before we expand the chain of inequalities any further, let us state two results: apply Lemma 4.5 with $a = 1/2$ to conclude that for any complex number z such that $|z| \leq 1/2$, we have $|\log(1+z)| \leq 2\log(2)|z| < 2|z|$. Moreover, for any nonzero complex number z,

$$|\log(z)| = |\log|z| + i\operatorname{Arg}(z)| = \sqrt{(\log|z|)^2 + (\operatorname{Arg}(z))^2} \geq \sqrt{(\log|z|)^2} = |\log|z||. \quad (5.5)$$

One consequence of $Y > Y_1$ is that $\left|\frac{X}{Y} - \theta^{(i)}\right| \leq \frac{1}{n+1}|\theta^{(i)} - \theta^{(i_j)}|$ (see proof of Proposition 2.2.1 (ii) in [10]), and in particular $\left|\dfrac{X/Y - \theta^{(i)}}{\theta^{(i)} - \theta^{(i_j)}}\right| \leq \dfrac{1}{n+1} \leq 1/2$. So now, from inequality (5.5), together with $|\log(1+z)| \leq 2|z|$ for $|z| \leq 1/2$, we see that $Y > Y_1$ implies

$$\left|\log\left|\frac{X/Y - \theta^{(i_j)}}{\theta^{(i)} - \theta^{(i_j)}}\right|\right| = \left|\log\left|1 + \frac{X/Y - \theta^{(i)}}{\theta^{(i)} - \theta^{(i_j)}}\right|\right| \leq \left|\log\left(1 + \frac{X/Y - \theta^{(i)}}{\theta^{(i)} - \theta^{(i_j)}}\right)\right| \leq 2\left|\frac{X/Y - \theta^{(i)}}{\theta^{(i)} - \theta^{(i_j)}}\right| \leq$$
$$\leq \frac{2}{\min\limits_{l \neq m}|\theta^{(l)} - \theta^{(m)}|} \cdot \left|X/Y - \theta^{(i)}\right| = c_1\left|X/Y - \theta^{(i)}\right|.$$

Lemma 7.3 in [4] implies that, if $c_9 \stackrel{\text{def}}{=\!=} \max\limits_{1\leq i\leq n}\left(\dfrac{2^{n-1}|m|}{\left|\frac{\partial F}{\partial X}(\theta^{(i)}, 1)\right|}\right)$, then $|\beta^{(i)}| \leq \dfrac{c_9}{|Y|^{n-1}}$. Further, $\dfrac{1}{|Y|} \leq c_{12}e^{-c_{13}A}$, hence $\dfrac{c_9}{|Y|^n} \leq c_9 c_{12}^n e^{-n\cdot c_{13}A}$ and thus

$$\left|\log\left|\frac{X/Y - \theta^{(i_j)}}{\theta^{(i)} - \theta^{(i_j)}}\right|\right| \leq c_1\left|X/Y - \theta^{(i)}\right| = \frac{c_1}{|Y|}|\beta^{(i)}| < \frac{c_1}{|Y|}\cdot\frac{c_9}{|Y|^{n-1}} \leq c_1 c_9 c_{12}^n e^{-nc_{13}A} = c_{14}e^{-c_{15}A},$$

where we define $c_{14} \stackrel{\text{def}}{=\!=} c_1 c_9 c_{12}^n$ and $c_{15} \stackrel{\text{def}}{=\!=} n \cdot c_{13}$.

Ultimately, after setting $c_{16} = c_{14}\max\limits_{1\leq k\leq r}\sum\limits_{j=1}^{r}|u_{k,j}|$, one obtains the inequality

$$|\delta_k \log|Y| - a_k + \lambda_k| \leq c_{16}e^{-c_{15}A}.$$

Next we set $|\delta_h| = \max\limits_{1\leq k\leq r}|\delta_k|$ and let g denote any index not equal to h. Further, set $\delta = \delta_g/\delta_h$ and let $\lambda = (\delta_g\lambda_h - \delta_h\lambda_g)/\delta_h$. At this step one may be hesitant of whether λ is well defined, i.e., $\delta_h \neq 0$. This turns out not to be an issue: $\delta_h = 0$ would force each row of U_I^{-1} to sum to 0. However, this forbids the existence of a matrix V with $UV = I$, thus contradicting the invertibility of U_I. With that remark aside, one can further show that

$$|a_g - \delta a_h + \lambda| \leq (1 + |\delta|)c_{16}e^{-c_{15}A} \leq 2c_{16}e^{-c_{15}A}. \quad (5.6)$$

At this stage, our attempt to determine an upper bound on A from inequality (5.6) translates to the problem of determining how small a nonzero linear form can be. Thus,

we can use inequality (5.6) together with the LLL-reduction techniques from Section 3.2 for the inhomogeneous linear form

$$\Lambda' = a_g - \delta a_h + \lambda,$$

in order to determine an upper bound on A. Note that approximation of the form is carried out using a two-dimensional lattice, irrespective of the degree of the underlying Diophantine problem. In this way we gain numerical efficiency in comparison to the method of Tzanakis and de Weger, especially when the degree of the Thue equation is rather large.

5.2 Final Enumeration

Suppose now that the reduction step gave us an upper bound $A \leq A_7$. Again, we have to iterate over all possible $(2A_7+1)^r$ vectors $(a_1, \ldots, a_r)$, together with the choices of roots of unity and representatives of norm $m \in \mathbb{Z}\backslash\{0\}$. It could however be that the reduced bound A_7 is too large for direct enumeration. We close this section with a suggestion on how to remedy such cases.

Define the constant $c'_{12} \stackrel{\text{def}}{=\!=} 2.78 \cdot \dfrac{c'_1}{c'_2}\|U_I^{-1}\|_\infty$, where c'_1 and c'_2 are given in Lemma 5.1. Then, let

$$c'_3 \stackrel{\text{def}}{=\!=} c'_1 \frac{n+1}{n} \cdot \frac{|\theta^{(m_1)} - \theta^{(m_2)}|}{|\theta^{(i)} - \theta^{(m_1)}| \cdot |\theta^{(i)} - \theta^{(m_2)}|}, \quad m_1, m_2 \neq i,$$

and $Y_2 \stackrel{\text{def}}{=\!=} \max(Y_1, (2c'_3)^{1/n})$.

Lemma 5.2. *For $1 \leq j \leq r$, let $a'_j = \dfrac{1}{\delta_h}(\delta_j a_h - (\delta_j \lambda_h - \delta_h \lambda_j))$. Suppose that $|Y| > Y_3 \stackrel{\text{def}}{=\!=} \max(Y_2, (2c'_{12})^{1/n})$. Then $|a_j - a'_j| < 1/2$.*

For details of the proof of Lemma 5.2, refer to Section 4.7 in [14] and Proposition 2.5.1 in [10]. The enumeration now proceeds as follows: assuming Y is large enough, that is $|Y| > Y_3$, we have $|a_j - a'_j| < 1/2$. In particular, a_j is the nearest integer to a'_j. Hence, each value a'_j uniquely determines a_j and so a vector $\vec{a'} = (a'_1, \ldots, a'_r)$ induces a unique vector $\vec{a} = (a_1, \ldots, a_r)$. The vector $\vec{a'}$ is determined once we make a choice for a_h. We have a total of $2A_7 + 1$ such choices for a_h, hence $2A_7 + 1$ vectors $\vec{a'}$, hence, just as many vectors $\vec{a}$. In turn, we need only loop through $2A_7 + 1$ possible cases for the exponents, in comparison to $(2A_7 + 1)^r$ possibilities from the method of Tzanakis and de Weger. This is yet another computational advantage for the method of Bilu and Hanrot, compared to the method of Chapter 4.

With this remark on the final enumeration, we close our discussion of the method of Bilu and Hanrot for solving Thue Equations.

6 Conclusion

Thue equations are an important class of Diophantine equations, which find several theoretical applications. A result, due to Axel Thue (1909) states that a Thue equation has only finitely many integer solutions. In the second half of the 20th century mathematicians developed methods for determining all integer solutions to an equation of this type. The first practical method is due to N. Tzanakis and Benjamin M. M. de Weger. In their approach, the classical Diophantine problem is reduced to the problem of finding all possible representations of elements of particular form inside a number field. Combining results by A. Baker on logarithmic forms, together with techniques from lattice reduction, Tzanakis and de Weger significantly reduce the number of possible representations, which allows for solving the Thue equation by direct enumeration.

We implemented the method of Tzanakis and de Weger in `Mathematica`. This allowed us to solve several fourth degree Thue equations. Our implementation has certain drawbacks in the step of final enumeration. Comparison with the method `Reduce` showed us that no solutions were lost during enumeration. However, this does not prove robustness of our implementation and leaves room for improvement of the code.

A system of fundamental units of a number field is unique up to coordinate transformation and multiplication by a root of unity. A special case of this transformation occurs when we replace a unit with its inverse. Our contribution to the thesis was in studying how different choices of fundamental units impact the performance of the method of Tzanakis and de Weger. We observed that choices of fundamental units can influence the upper bounds for the exponents of the units in the representations of desired form. This notice allowed us to suggest a strategy for a choice of units which will improve computational efficiency. One possible future consideration could be studying a more general $GL_n(\mathbb{Z})$ action on the system of fundamental units.

One modification of the method of Tzanakis and de Weger is found in a method due to Yuri F. Bilu and Guillaume Hanrot. The latter method gives us certain numerical advantages. Bilu and Hanrot solve the problem of lattice approximation using only two-dimensional lattices. In their approach, the number of cases during the final enumeration is further reduced. This is makes the method more efficient, especially when the degree of the underlying Thue equation is large.

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